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Traversing the storm: An interdisciplinary review of crisis leadership

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ABSTRACT

An organizational crisis is a low-probability, high-impact event that threatens the survival of organizations and individuals, often with little warning. In response, people seek clarity, reassurance, and hope from organizational leaders. Yet, crises also vary in nature and impact (e.g., a product failure versus the COVID-19 pandemic), which presents diverse challenges to leaders and differing stakeholder perceptions. Based on a critical analysis of 69 empirical articles, we provide a comprehensive, systematic, interdisciplinary review of the crisis leadership literature. Our review utilizes the Coombs and Holladay (1996) crisis typology, where crises are categorized according to mutually exclusive attributional dimensions (i.e., internal–external and intentional–unintentional). We conduct a thematic analysis of crisis leadership within and across these four crisis categories and find that each is associated with a different leadership theme. We also examine the methodological quality and rigor of the qualitative and quantitative articles in our review. Based on our findings, we also offer suggestions to guide future crisis leadership research, and provide guidance for organizational leaders in how to respond to various crises.

Introduction

“Anyone can hold the helm when the sea is calm” (Publius Syrus, 85–43 BC)

When faced with a crisis, those impacted look to their leaders for direction, reassurance, and hope (James & Wooten, 2010). Crises are urgent, ambiguous, and high stakes events, that have the potential to exhaust resources and destroy the reputation of individuals and organizations (James, Wooten, & Dushek, 2011). For leaders, responding to a crisis is complicated because any type of crisis forces leaders to deal with challenges that are very different from everyday business problems. For example, the COVID-19 pandemic, a crisis that has wreaked havoc on organizations worldwide and is to date responsible for over four and a half million deaths (International Monetary Fund, 2021; World Health Organization, 2021), is an event that most leaders were not prepared to handle (Al Saidi et al., 2020; Crayne & Medeiros,

2021). While crises were once considered to be relatively rare (Pearson & Clair, 1998), they are predicted to increase in frequency due to climate change, population growth, and rising geographic density (World Economic Forum, 2020). As such, leaders will face the daunting prospect of dealing with more frequent, more intense, and more damaging crises than ever before. Given these dire forecasts, a review of crisis leadership is both important and timely for three reasons.

First, despite the critical role of leadership during a crisis, some leaders may lack the confidence or ability to respond effectively (Malinen, Hatton, Naswall, & Kuntz, 2019). Crisis leadership is complicated by the fact that leadership actions desired by stakeholders will vary according to the different types of crises being faced. For example, North et al. (2013) found that leaders felt ill-equipped to deal with the magnitude of employees' emotional responses following the 9/11 terrorist attack. More recently, a poll of leadership perceptions during

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COVID-19 found that only 39 % of employees in the US strongly agreed that their organization had communicated a clear action plan, while less than half strongly agreed that their leader kept them informed of the organizational impact of the pandemic (Harter, 2020). Given these concerns, a review of the empirical crisis leadership literature can provide both scholars and practitioners with evidence-based insights into effectively leading organizations during different types of crises.

Second, given the ongoing impact of the COVID-19 pandemic, there is growing scholarly interest in understanding the critical role of leadership during crises (see Fig. 1). There have been recent editorials in relation to crisis leadership (e.g., Forster, Patlas, & Lexa, 2020; Rubin & Hamel, 2020; Spalding, Karlsson, Hirschmann, Becker, & Musahl, 2020); special issues and calls for papers (e.g., Jacquart, Santoni, Schudy, Sieweke, & Withers, 2020; Kim, Vaiman, & Sanders, 2020; Mau & Ohemeng, 2020; Newman, Eva, Bindl, & Stoverink, 2020; Tourish, 2020); and empirical and conceptual pieces published (e.g., Dasborough & Scandura, 2021; Hu, He, & Zhou, 2020; Rudolph et al., 2021). In addition, *The Leadership Quarterly* published a recent bibliometric review of crisis leadership that mapped out linkages in the extant crisis leadership literature (Bavik, Shao, Newman, & Schwarz, 2021).

Despite Bavik et al. (2021) bibliometric analysis, which brought together various fragmented streams of crisis leadership literature, our understanding of how leaders respond to specific types of crises remains underdeveloped. “Type of crisis” refers to the particular crisis situation being faced (Coombs & Holladay, 1996); for example, crises include a wide range of damaging events such as floods, bushfires, wars, terrorism, product failures, ethical scandals, global pandemics, and global financial crises. We aim to build on Bavik et al. (2021) recent contribution by exploring how different types of crises may require different responses by leaders, answering Bavik et al. (2021) call to study leadership across crisis contexts. Crises of different kinds have unique properties and pose various challenges to organizational leaders (James et al., 2011). Therefore, examining leadership within and across different crisis types is important (Bavik et al., 2021). Despite the existence of various crisis typologies, few crisis leadership studies utilize them (e.g., Bavik et al., 2021). To address this, we draw on an existing crisis typology (Coombs & Holladay, 1996) to explore leadership in response to various types of crises. By identifying common themes that emerge from the crisis leadership literature, we hope to provide valuable recommendations for future research and leadership practice in response to different types of crises.

Finally, there has been little assessment of the empirical crisis leadership literature in terms of its quality or methodological rigor. *The Leadership Quarterly*’s mission is to publish studies of high methodological quality, and critically assessing the crisis leadership literature allows us to identify areas for methodological improvement. Assessing the quality of crisis leadership research is important because empirical research that lacks methodological rigor hinders the ability of management scholars to provide meaningful advice with real world policy implications (Antonakis, 2017). While the review by Bavik et al. (2021) lists the empirical designs used and suggests that natural experiments should be employed to study crisis leadership, their review offers little in terms of critiquing the methodologies utilized in prior studies of crisis leadership. Here, we contribute to the crisis leadership literature by identifying methodological flaws in existing empirical studies of crisis leadership and provide recommendations for future qualitative and quantitative research following established guidelines (Aguinis & Solarino, 2019; Antonakis, 2017; Antonakis, Bendahan, Jacquart, & Lalive, 2010; Hill, Johnson, Greco, O’Boyle, & Walter, 2021).

Before we address these aims, we begin our review by clarifying definitional issues that have hindered crisis leadership research. It is important to clearly define what an organizational crisis is (and is not), and to define crisis leadership. We also provide our rationale

for using the Coombs and Holladay (1996) crisis typology to classify crisis leadership research. Once these foundational conceptual arguments are made, we then explain the systematic search method employed to identify the studies included in this review.

Definitions and conceptual foundations

Definitions: Organizational crisis and crisis leadership

Pearson and Clair (1998) define an organizational crisis as “a low-probability, high-impact event that threatens the viability of the organization and is characterized by ambiguity of cause, effect, and means of resolution, as well as by a belief that decisions must be made swiftly” (p. 60). Drawing on this definition, and following earlier work by Hermann (1963), James et al. (2011) argue that organizational crises are distinct from “run-of-the-mill business challenges”³ due to three key elements: “rarity of the event, significance of the event, and the level of impact on stakeholders” (p. 461). First, an organizational crisis must be a *rare, abnormal, or infrequent event*. This means that the organization has little or no prior experience in managing the event, or there are few if any well-rehearsed ways to deal with the crisis. Second, it must pose a *significant threat to the organization’s long-term health and viability*. That is, the crisis itself, or mismanaging the crisis, can potentially exhaust resources and destroy the organization. Finally, it must have the potential to result in a *profoundly negative impact on stakeholders*. Stakeholders who may be impacted by the crisis include people within the organization (e.g., employees), as well as people outside of the organization (e.g., customers, suppliers, media, the public).

Bavik et al. (2021) conceptualized crisis leadership as “a leadership process around times of crisis, including times immediately prior to crises, the duration of crises as they unfold, and times immediately after the acute consequences of crises” (p.3). Building on this definition of crisis leadership and the earlier definition of organizational crises discussed previously, we propose that crisis leadership is “*the influencing process that occurs between a leader and stakeholders (internal and external) in the context of organizational crises, as opposed to run-of-the-mill business challenges, over the various stages of the crisis lifecycle*”.⁴ Bavik et al. (2021) explain that crisis leadership focuses on how leaders influence different stakeholders during crises, why some leaders are more effective than others in crisis contexts, and how leaders are impacted by the crisis themselves. In our review, we examine these elements of crisis leadership and we contrast how leaders respond to different types of crises.

Much of the scholarly research on leaders during an organizational crisis has focused on either *crisis leadership* or *crisis management*. While in the general leadership literature there are notable distinctions in behaviors of “leaders” who inspire, motivate, and guide and “managers” who budget, hire, and supervise, Kniffin, Detert, and Leroy (2020) found that there is a significant overlap between the two roles. Based on their findings, they call for leadership scholars to consider both leading and managing activities simultaneously.

Indeed, navigating through a crisis requires elements of both leadership and management, and the terms “crisis management” and “crisis leadership” are often used interchangeably (James et al., 2011). Boin, Hart, McConnell, and Preston (2010) argue that “effective crisis leadership entails recognizing emerging threats, initiating efforts to mitigate them and deal with their consequences, and once an acute cri-

³ James et al. (2011) argue that a train derailment, for example, is a business challenge that would only qualify as a crisis in certain situations, e.g., if it were carrying toxic chemicals that unexpectedly leaked and threatened the health of people or ecosystems near the derailment.

⁴ We also distinguish our review of crisis leadership from leadership in extreme contexts (e.g., Hällgren, Rouleau, & de Rond, 2018) and emergency management (e.g., Oh & Lee, 2020), as these are not “organizational” crises contexts in terms of our definition.

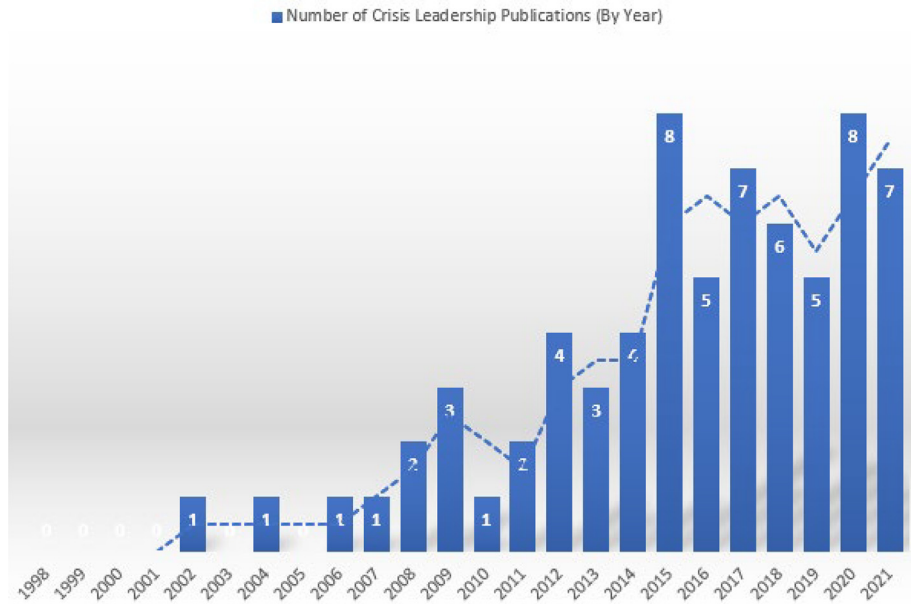


Fig. 1. Upward trend in crisis leadership research. Note: This graph only includes the empirical articles selected for inclusion in our review.

sis period has passed, re-establishing a sense of normalcy” (p. 706). Hence, effective crisis leadership involves aspects of managing (e.g., planning and organizing), while also leading the crisis response of internal and external stakeholders. The actions of organizational leaders therefore involve an “internal” orientation, focused on managing the technical and structural elements of the crisis itself, and an “external” orientation focused on managing stakeholder perceptions (Bundy, Pfarrer, Short, & Coombs, 2017). As such, in this review of crisis leadership, we examine what leaders do, both in terms of the actions they take to manage the effects of a crisis, and how they serve as leaders during a crisis.

Classification of crises - typologies

The recognition that different types of crises can cause distinct stakeholder responses has led scholars to propose various typologies based on crisis characteristics (e.g., Marcus & Goodman, 1991; Williams, Gruber, Sutcliffe, Shepherd, & Zhao, 2017). Crisis typologies serve as a means of sensemaking (Weick, 1988) as they reduce the ambiguity associated with a crisis event by providing a framework for taking appropriate action (James et al., 2011). In this review, we utilize a theory-driven approach by drawing on the Coombs and Holladay (1996) crisis typology to categorize empirical studies of different crisis types.

The Coombs and Holladay (1996) crisis typology provides a parsimonious way to categorize crisis events and relate crisis leadership to various internal and external stakeholder perceptions of the leader across different crisis types. Unlike some other typologies (e.g., Rosenthal & Kouzmin, 1997), this typology encompasses all the types of crises identified in our review. Further, this typology meets all of Gundel's (2005) criteria for a useful crisis typology, namely, that: (1) the categories are mutually exclusive, (2) the model is exhaustive, (3) the categories are relevant, and (4) the model is pragmatic (see also James et al., 2011).

One of the main attractions of the Coombs and Holladay (1996) crisis typology is that it draws on attribution theory (Heider, 1958; Weiner, 1986) to classify crises as events that originate either externally or internally to the organization and arise either intentionally

or unintentionally.⁵ Crises caused by events that are *external* to the organization and *intentional* include acts of terrorism, such as the 2011 Oslo bombing (Birkeland, Nielsen, Hansen, Knardahl, & Heir, 2017), whereas extreme weather events, such as Hurricane Katrina in 2005 are categorized as *external* and *unintentional* events, as these events do not directly arise from deliberate human acts (Smith, Sabbag, & Rohmer, 2018). Crises can also be caused by events that are *internal* to the organization and can similarly be classified as either *intentional* or *unintentional*. The Enron scandal of 2001 is an example of a crisis caused by *intentional* acts that were *internal* to the organization (Hannah & Zatzick, 2008). *Internal* crises can also arise from acts that are *unintentional*; for example, Roche's 2007 recall of its Viracept AIDS medication due to contamination (Bauman, 2011).

By drawing on attribution theory (Weiner, 1986), this typology can provide useful insights into crisis leadership, as leader behaviors can alter stakeholders' causal attributions (Martinko, Harvey, & Dasborough, 2011). Attributions of leader behaviors are important, as they shape stakeholder emotions and behaviors, and ultimately determine stakeholder endorsement of leaders (Dasborough, Ashkanasy, Tee, & Tse, 2009). Therefore, to aid our understanding of leadership across different crisis contexts, all the studies we include in this review were classified according to the Coombs and Holladay (1996) crisis typology.

Method

Article Search, Inclusion, and coding

We undertook a comprehensive and systematic search of the literature to identify qualitative and quantitative studies to include in our review of crisis leadership. To conduct our search, we followed best practices in the field (e.g., Parmigiani & King, 2019; Short, 2009; Siddaway, Wood, & Hedges, 2019; see Fig. 2). Using Pearson and Clair's (1998) seminal crisis article as the starting point, our search

⁵ Coombs and Holladay (1996) address the three causal dimensions from attribution theory in their typology, i.e.: locus (internal/external), controllability (intentional/unintentional), and stability. Although stability is not explicit in their typology, they explain how “each crisis type can vary along the stability dimension” (p. 285).

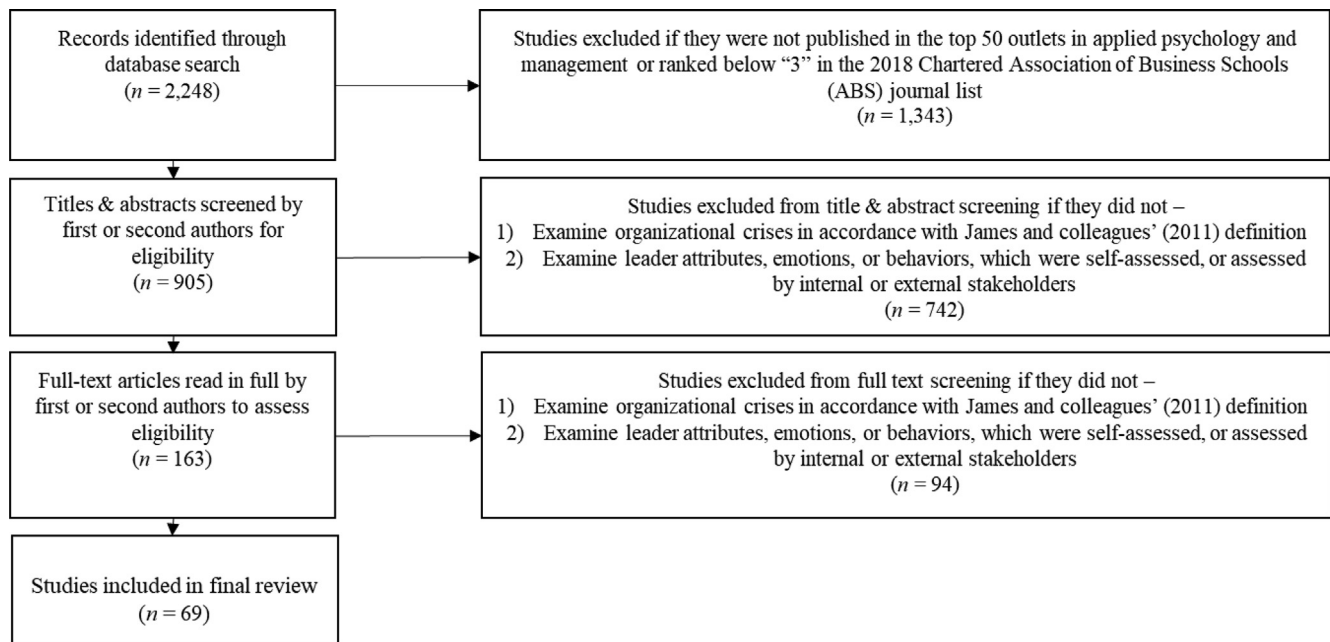


Fig. 2. Literature search process.

covers the period between 1998 and July 2021. We searched the PsycInfo and Web of Science databases using crisis-related keywords (crisis, disaster, shock, jolt, turbulence, and scandal) and leadership-related keywords (leader*, manage*, and supervis*) to identify relevant articles. After removing duplicates, this process yielded $k = 2,248$ articles for potential inclusion in our review.

We reviewed only high-quality empirical research, restricting articles from this initial pool to those published in the top 50 outlets in applied psychology and management (e.g., Antonakis et al., 2019), and those ranked “3” and higher in the 2018 Chartered Association of Business Schools (ABS) journal list (e.g., Castro, Phillips, & Ansari, 2020). We also included articles from specialist journals that specifically target crisis, disaster, and/or risk management (e.g., *Disaster Prevention and Management*, *Journal of Contingencies and Crisis Management*) as we were mindful of the need to identify as many relevant articles as possible (Siddaway et al., 2019), while also minimizing random sampling (Short, 2009). This led to the removal of a further $k = 1,343$ articles, leaving $k = 905$ articles for potential inclusion in our review.

In contrast to the recent review by Bavik et al. (2021), which used computer-assisted bibliometric analysis,⁶ the first and third authors applied strict a priori inclusion criteria when reading all 905 remaining abstracts. To be included in the review, articles had to examine an organizational crisis that was: (1) rare, (2) a significant threat to the organization’s survival, and (3) posing a negative impact on stakeholders (James et al., 2011). Articles that did not meet these criteria were excluded, including those examining what appeared to be similar constructs, such as: environmental “uncertainty” (e.g., Waldman, Ramirez, House, & Puranam, 2001), those describing a “crisis”, but not meeting all three elements of the definition (e.g., common business challenges, Flynn & Staw, 2004), and those examining workplace “stress and anxiety” (e.g., Pillai & Meindl, 1998), or “task load” and “time constraints” (e.g., Steinrücke, Veldkamp, & de Jong, 2019), as proxies for organizational crises.

In addition, drawing on Antonakis, Cianciolo, and Sternberg (2004), all articles had to examine leadership. This could include leader attributes, emotions, or behaviors, which were self-assessed, or

assessed by internal or external stakeholders (e.g., followers, higher-level managers, voters).⁷ Finally, in contrast to Bavik et al. (2021), only empirical studies were included in our review; non-empirical, prescriptive, and conceptual articles (e.g., Heyden, Wilden, & Wise, 2020; Kooor-Misra, 2002; Probst & Raisch, 2005) were excluded as we wanted our review and proposed guidelines for future research to be evidence based.

To ensure consistency in applying the inclusion criteria described, the first and third authors independently read abstracts of the same 10 articles to assess agreement while on a Zoom call. Discussion during this training aimed to maximize alignment in the way these authors made determinations about inclusion versus exclusion. Next, to assess the efficacy of this training, the first and third authors independently coded the same 92 articles (approximately 10 % of all remaining articles). They had agreement on 92 % of these articles (Cohen’s kappa = 0.74), suggesting a relatively high level of inter-rater reliability. The authors discussed the small number of inclusion or exclusion discrepancies and found that these differences were mostly due to abstracts that were unclear or ambiguous. In these cases, the full article was read in detail and discussed, after which a decision to include or exclude was made. Following this training session, the authors equally divided the 802 remaining articles and independently coded them to decide if they should be included or excluded. A total of $k = 742$ articles were excluded at this stage, leaving $k = 163$ articles retained for potential inclusion.

The lead authors then independently read each of the remaining $k = 163$ articles in full, again paying close attention to the a priori inclusion criteria and following the aforementioned procedure for dealing with discrepancies. Reading the full text of the articles revealed that some, although initially included based on the abstract, did not actually meet the a priori inclusion criteria (i.e., they did not focus on a rare event that threatened organizational survival and negatively impacted stakeholders; they did not focus on leadership; or they were not empirical). A further $k = 94$ articles were excluded at

⁶ Where inclusion was based on co-citation, bibliographic coupling, and co-word analysis.

⁷ We excluded studies that did not examine the attributes, behaviors or emotions of focal leader(s), e.g., binary measures like CEO replacement (e.g., Gomulya & Boeker, 2016).

this stage resulting in the final sample of $k = 69$ articles retained for the review.

Next, the lead authors independently coded all $k = 69$ articles. Like the process described to assess the inclusion/exclusion of articles, prior to coding all the articles, the lead authors independently read 10 of the same articles and coded them while on a Zoom call. Discussion during this training ensured alignment in the way both authors coded all 69 articles. For each study a description of the crisis investigated, the specific leader/s involved, the leader behaviors/styles employed (e.g., leader used charismatic rhetoric), and key findings were noted. Subsequently all studies were then coded by crisis type according to the [Coombs and Holladay \(1996\)](#) typology (e.g., external vs internal, intentional vs unintentional), for leader level (e.g., frontline leader, strategic level leader), for leadership or related theories employed (e.g., transformational, authentic leadership), and for study design (i.e., the study was quantitative, qualitative, used mixed methods).⁸ In addition, all studies were also coded for methodological rigor, guided by current best practices in qualitative and quantitative research (e.g., [Aguinis & Solarino, 2019](#); [Antonakis, 2017](#); [Hill et al., 2021](#)). See [Tables 1 and 2](#) for coding categories, and [Table 3](#) for details of the leadership studies in each of the crisis type quadrants. The lead authors checked their relative coding on a sub-sample of 33 studies. The coding agreement on these 33 studies was very high at 96 % (Cohen's kappa = 0.85).

The empirical crisis leadership literature – Study characteristics

Leaders at various levels were featured in the studies of crisis leadership included in this review. The most common types of leaders examined were executive level organizational leaders ($k = 35$, 50.72 %) and political leaders ($k = 19$, 27.53 %). Other studies examined frontline or middle-level managers ($k = 7$, 10.14 %) (e.g., [Post, Latu, & Belkin, 2019](#); [Sommer, Howell, & Hadley, 2016](#); [Zhuravsky, 2015](#)). The rest of the articles investigated general leadership, either not giving specific details about the types of leaders studied or studied leaders across organizational levels ($k = 8$, 11.59 %). There was no discernable pattern in the types of leaders studied for three out of the four [Coombs and Holladay \(1996\)](#) crisis types. The only exception to this was the studies of internal-unintentional crises, where all but one study focused on executive level leaders (CEO's). With regards to leader gender, the overwhelming majority of articles focused on male leaders. Only nine studies that focused on the role of gender examined women leaders. Almost half of these were in the external-unintentional category, and the rest were in the internal-unintentional category. Of note, in the internal-intentional quadrant leaders were often viewed as the cause of the crisis, and in an overwhelming majority of cases the leader was male.⁹

Thirty-two of the 69 crisis leadership articles reviewed adopted quantitative methodologies (46.38 %), with 10 of those (31.25 % of all quantitative studies) employing experimental designs and 22 sourcing their data from either field surveys ($k = 8$, 25 % of quantitative studies) or archival information ($k = 14$, 43.75 % of quantitative studies). There were also 30 qualitative studies (43.48 % of all studies), and seven mixed method studies (10.14 % of all studies) included in the review. Different methodologies were seen in each quadrant; however, in the external-unintentional quadrant the majority of studies were either qualitative or employed mixed methods, while in the other three quadrants the studies were mostly quantitative. Another interesting observation was that experiments employing vignettes featured prominently in the internal-unintentional quadrant which focused on

CEO's. These trends in methodology by crisis type were an unexpected finding from our review.

Only around a third of the empirical studies in our crisis leadership review are based on established leadership theories. Charismatic leadership theory ($k = 6$, 8.70 %) and transformational leadership theory (including the full range leadership model, $k = 3$, 4.35 %) have been examined within the crisis context. Also, since leaders are expected to do the right thing in crises situations, ethical leadership theory ($k = 7$, 10.14 %) and authentic leadership theory ($k = 2$, 2.90 %) were also utilized to explain leader behaviors in response to a crisis. Other theoretical perspectives employed were supportive leadership theory ($k = 2$, 2.90 %), shared leadership theory ($k = 1$, 1.45 %), leader-member-exchange theory ($k = 1$, 1.45 %), trait theory of leadership ($k = 4$, 5.80 %), and implicit leadership theory ($k = 1$, 1.45 %). In addition to the different leadership styles, two other theoretical foundations have been widely adopted in this literature. Theories regarding top management teams (e.g., upper echelon theory, $k = 10$, 20.29 %) reflect the emphasis on strategic level leaders in the current literature, and leadership and gender ($k = 10$, 14.49 %) has also been of interest to scholars of crisis leadership. The only pattern in terms of theory across quadrants is that ethical leadership theory features in the internal-intentional quadrant, and to a lesser extent in the internal-unintentional quadrant.

Themes emerging from the review

Using an inductive process, we were able to identify the themes that emerged from the empirical crisis leadership literature by consolidating the common and salient findings within each quadrant of the [Coombs and Holladay \(1996\)](#) crisis typology. Specifically, we followed the recommended iterative process to arrive at the core themes for each quadrant, as follows: (1) gaining familiarity with the data by deeply reading all included studies; (2) generating initial codes and organizing them into meaningful categories (e.g., leader behavior or style, study focus, key findings); (3) searching for candidate themes that emerged from an analysis of the initial code categories; (4) reviewing candidate themes and referring back and forth to the studies to finalize core themes; (5) defining and naming the themes along with their supportive contents; and finally (6) writing up the findings ([Braun & Clarke, 2006](#); [King & Brooks, 2018](#)). This process enabled us to consolidate and make meaning from the literature reviewed by identifying the different leadership behaviors that have been empirically studied for each type of crisis.

The themes that emerged from our review were as follows: (1) external-unintentional crises - *leaders as shepherds*; (2) external-intentional crises - *leaders as saints*; (3) internal-unintentional crises - *leaders as spokesperson*; and (4) internal-intentional crises - *leaders as sinners*. These themes emerged from the common findings within each quadrant and are descriptive, in that they reflect what leaders have been observed to do empirically in response to each type of crisis. Next, we review the findings of empirical studies within each quadrant that led to the identification of these themes, and then highlight similarities across crisis types.

External crises

(1) Leaders as shepherds: Protecting and guiding through external-unintentional crises. Over half the crisis leadership articles in this review examined external-unintentional crises ($k = 41$, 59.4 %). The studies in this quadrant can be divided into two sub-categories: studies that investigated industry-wide financial shocks, which are unintentional crises arising from human actions ($k = 22$), and studies of crises caused by natural disasters ($k = 19$), i.e., "acts of god" including hurricanes, floods, earthquakes, and epidemics/pandemics. Leader responses to financial shocks typically involved cen-

⁸ See [Table 1](#). We also coded each article for sample type, sample size, year published, country in which data was collected, and the method used to collect data. None of these variables showed notable patterns that affect the themes that emerged from our review.

⁹ In the study about the Enron bankruptcy, [Hannah and Zatzick \(2008\)](#) noted that their analysis had very few portrayals of female leaders. They tried to oversample women, but there were very few to be selected in the magazine articles they coded.

Table 1

Coding guide for crisis leadership themes.

Code	Code Description	Examples
Crisis Type (categories based on Coombs & Holladay, 1996).	a. External and unintentional	2010 Pakistan floods (Akbar & Aldrich, 2015); Financial shock, 2001 Ansett collapse (Bonn & Rundle-Thiele, 2007); 2008 GFC (Buyl et al., 2019)
	b. Internal and unintentional	Product harm crisis, 2007 Viracept recall (Bauman, 2011); Product harm crises, OTC pharma products (Cowen & Montgomery, 2020); School fire in Sweden (Danielsson & Sjöstedt-Landén, 2020)
	c. Internal and intentional	2009 British MP expense scandal (Graffin et al., 2013); Sex scandals (Grover & Hasel, 2015); Corporate scandals, e.g., Enron, WorldCom (Hannah & Zatzick, 2008)
	d. External and intentional	2011 Oslo bombing (Birkeland et al., 2017); 9/11 attacks (Bligh et al., 2004); 2006 Lebanon War (Boehm et al., 2010)
	e. Multiple crisis types	9/11 attacks, Hurricane Katrina (Davis & Gardner, 2012); War, national deficit, terrorism, and foreign relations (Williams et al., 2009)
Leader Level	Political leaders	Municipal chief officers and politicians (Enander et al., 2009); US Governors (Jensen et al., 2021); 2008 US Presidential candidates, Obama & McCain (Williams et al. (2012)
	Executive level leaders	CEOs (Gorn et al., 2008; Knights & McCabe, 2015; Liu, 2017); Board members (von Walter et al., 2016)
	Frontline/middle level managers	Nurse unit managers (Sommer et al., 2016); Production team leaders (Post et al., 2019); Hospital managers (Zhang et al., 2012)
	Leaders at mixed levels	US higher education leaders (Liu et al., 2021); Nurse/clinical managers (Zhuravsky, 2015); US Presidents, professors, & CEO (Stam et al., 2018)
Theoretical Foundation / Perspective	Attribution Theory Authentic Leadership Charismatic Leadership Ethical Leadership Full Range Leadership Model Gender and Leadership Leadership Trait Theory Supportive Leadership Transformational Leadership Top Management Teams Upper Echelon Theory Other None	Steinbach et al. (2021)Liu et al. (2017)Jensen et al., (2021)Grover and Hasel (2015)Sommer et al. (2016)Elliott and Stead (2018)Van Scotter and Roglio (2020)Birkeland et al. (2017)Zhang et al. (2012)Klein and Eckhaus (2017)Gupta et al. (2018)Stam et al. (2018)Gorn et al. (2008)
Study Design	Qualitative	Discourse analysis (Liu, 2017; Liu et al., 2017); Case study (Knights & McCabe, 2015)
	Observational (archival)	Gupta et al. (2018); Kashmiri and Brower (2016); Patel and Cooper (2014)
	Observational (survey)	Akbar and Aldrich (2015); Davis and Gardner (2012); Sommer et al. (2016)
	Experimental vignette	Cowen and Montgomery (2020); Laufer et al. (2018); Jensen et al. (2021)
	Mixed-method	Bligh et al. (2004); Sergeant and Stajkovic (2020); Varma (2020)

tralizing authority, making strategic decisions more quickly, and being more intuitive to ensure their organizations survived (e.g., Bonn & Rundle-Thiele, 2007). In response to “acts of god”, leaders were concerned with protecting the lives of their followers through being adequately prepared, taking steps to minimize damage, and assisting recovery (e.g., Hede, 2011). In both cases, leaders were observed to provide a sense of safety and direction to move forward and recover from the crisis event.

Accordingly, the overarching theme in this quadrant reflected *leaders as shepherds*, namely, protecting their organizations and their followers and guiding them to safety. For leaders in this quadrant, “shepherding” came with many unexpected challenges that potentially worsens the impact of a crisis when mismanaged (e.g., Akbar & Aldrich, 2015; Hsu, 2017). Effective crisis leaders appeared to demonstrate a number of critical actions over the course of the crisis lifecycle. First, they identified the early warning signs of an impending crisis

(e.g., Boin, Hart, Stern, & Sundelius, 2005; Enander, Hede, & Lajksjö, 2009). Next, they took steps to prevent and/or constrain the damage caused by a crisis (e.g., by planning and preparing for various crisis eventualities, Steinbach, Kautz, & Korsgaard, 2021; Swinkels, 2020). Third, stakeholders looked to leaders to guide the recovery process following a crisis (e.g., Buyl, Boone, & Wade, 2019; Hede, 2011). Finally, leaders facilitated learning after the crisis had passed to prevent or prepare for similar crises in the future (e.g., Hede, 2011; Tourish & Hargie, 2012; Wooten & James, 2008). These actions by leaders all involved making effective decisions and helping both internal and external stakeholders understand what was happening and what needed to be done before, during, and after the crisis (Hsu, 2017; Mora Cortez & Johnston, 2020).

Leading through this type of crisis requires quick, rational, and ethical decision-making (Mora Cortez & Johnston, 2020). To protect their organizations and their people, leaders facing external-unintentional

Table 2

Coding guide for crisis leadership methodological critique.

Code	Criteria	Codes Utilized
Qualitative Research – Transparency (source of criteria and scale for coding: Aguinis & Solarino, 2019)	1. Kind of qualitative method	0 = Criterion not mentioned
	2. Research setting	1 = Criterion mentioned but not elaborated
	3. Position of researcher along the insider–outsider continuum	2 = Criterion partially met
	4. Sampling procedures	3 = Criterion met
	5. Relative importance of the participants/cases	
	6. Documenting interactions with participants	
	7. Saturation point	
	8. Unexpected opportunities, challenges, and other events	
	9. Management of power imbalance	
	10. Data coding and first-order codes	
	11. Data analysis and second- or higher-order codes	
	12. Data disclosure	
Qualitative Research – Coding	13. Coding reliability and validity (Wright, 2017)	0 = Criterion not mentioned 1 = Criterion mentioned but not elaborated 2 = Criterion partially met 3 = Criterion met
	14. Ensuring counterfactual evidence (Geddes, 2003; Gerring & McDermott, 2007).	
Quantitative Research – Threats to Methodological Rigor (source of criteria and scale for coding: Antonakis et al., 2010)	1. Omitted variables	0 = Irrelevant criterion
	2. Omitted selection	1 = Relevant criterion for which the authors did not correct
	3. Simultaneity	2 = Relevant criterion for which we were unable to determine whether it was taken into account by the authors
	4. Measurement error	3 = Relevant criterion which the authors addressed
	5. Common-method variance	
	6. Inconsistent inference	
	7. Model misspecification	

crises change the way they do things, as crises cause disruptions that alter the normal rhythms of organizational life (Williams et al., 2017). Crisis leaders move from making choices via a regular clock and calendar time (doing what is scheduled), to taking command and doing what is needed now by immediately responding to the crisis (Bonn & Rundle-Thiele, 2007; Mutch, 2015; Williams et al., 2017). This is because leaders who delay making important decisions during an external-unintentional crisis can potentially harm the lives and/or livelihood of their people, as well as the survival of their organization (e.g., Akbar & Aldrich, 2015; Rule & Tskhay, 2014). As such, decisions are made quickly and intuitively, often “on the fly”, based on less in-depth analysis, limited consultation, and by considering fewer strategic alternatives (e.g., Bonn & Rundle-Thiele, 2007; Enander et al., 2009; Liu et al., 2021). Importantly, leaders are often the target of considerable criticism when their decisions are deemed to be wrong and when they are seen to mismanage the crisis response (e.g., Akbar & Aldrich, 2015; Hsu, 2017; Rule & Tskhay, 2014; Smith et al., 2018).

The studies in this quadrant suggest that effective decision making by crisis leaders first involves making sense of the crisis (Boin et al.,

2010). In this regard, leaders have a “sensemaking” role, not only for themselves, but also for their internal and external stakeholders (e.g., Enander et al., 2009; Mutch, 2015; Windsor, Dowell, & Graesser, 2014). Importantly, sensemaking appears to underscore effective decision-making during a crisis, where decisions are being made under pressure (Boin & Renaud, 2013). Indeed, decision-making under pressure is considered a hallmark of effective crisis leadership (Boin et al., 2010), underscoring a leader’s credibility when directing, convincing, and coaching others throughout and following the crisis (e.g., Enander et al., 2009; Hsu, 2017; Jong, Dückers, & Velden, 2016). Even in crisis conditions that are fast-moving and frightening, leaders who make sense of a crisis for themselves and others appear to create a powerful regenerative effect, enabling renewal and energizing restorative collective action (Mutch, 2015; Riaz, Buchanan, & Ruebottom, 2016; Windsor et al., 2014).

Finally, we found that effective leadership during an external-unintentional crisis can also rely on a leader’s source of personal power, such as charisma, confidence, and empathy. For example, Jensen et al. (2021) found that charismatic US governors had lower death rates during the COVID-19 pandemic. Similarly, another study of US governors’ responses to the pandemic found that female US governors displayed more empathy and confidence than their male counterparts and had fewer deaths on average (Sergent & Stajkovic, 2020). In response to the 2008 global financial crisis (GFC), confident leaders were seen to be more authentic (Liu, Cutcher, & Grant, 2017). While leader charisma, confidence, and empathy appeared to be associated with effective crisis leadership across all crisis types, these sources of personal power were especially prevalent when responding to external-unintentional crises, where internal and external stakeholders are seeking guidance from leaders who appear to be in control of the crisis.

(2) Leaders as saints: Providing a beacon through external-intentional crises. Studies examining external-intentional crises ($k = 8$, 11.59 %) investigated leadership in response to war and terrorist acts. Whether they were political leaders, community leaders, or frontline supervisors, leaders studied in the context of external-intentional crises were seen to support and ultimately empower their followers to successfully navigate the crisis (e.g., Birkeland et al., 2017; Bligh, Kohles, & Meindl, 2004; Boehm, Enoshm, & Michal, 2010; North et al., 2013). Leaders facing an external-intentional crisis were seen to be more effective when they emphasized a collective focus, ensured continuity between the past and present, and shared a vision of overcoming adversity by rallying against a common enemy. Empowering others is a way that leaders can help followers return to a sense of normalcy (Birkeland et al., 2017).

Accordingly, the major theme that we derived from this quadrant reflects *leaders as saints*, that is, acting as a beacon of empowerment and support for followers negatively affected by external crises caused intentionally by others. The leaders studied in this crisis context not only had to respond to the aftermath of an initial crisis caused by malevolent intent (e.g., 9/11 or the 2011 Oslo bombing), but also faced situations where the threat of subsequent crises caused by the perpetrators were omnipresent (e.g., Bligh et al., 2004; Boehm et al., 2010; North et al., 2013). This makes leadership in response to an external-intentional crisis extremely difficult because both leaders and followers may feel highly fearful about their future (Davis & Gardner, 2012; North et al., 2013).

Bolstering the theme of sainthood, *leaders are revered* when they provided support for victims of external-intentional crises. For example, Birkeland et al. (2017) found that employees both noticed and appreciated the improvements in their immediate supervisor’s levels of supportive and empowering leadership behaviors immediately following their return to work after the 2011 Oslo bombings. Similarly, North et al. (2013) found that following the 9/11 terror attacks, employees attributed their own emotional recoveries to their leader’s supportive actions.

Table 3

Crisis leadership quadrants.

Author(s)	Crisis Description	Leader level (and Description)	Leadership or Related Theory	Leader Behavior or Style	Study Design	Key Findings
a. External and unintentional crises						
Akbar and Aldrich (2015)	2010 Pakistan floods	Political leaders (n = unknown)	Active leadership	Leader accessibility; effort; connectivity (to national/international NGOs); and sympathy	Quantitative (survey)	Flood victims equated higher levels of damage to ineffective crisis leadership.
Bonn and Rundle-Thiele (2007)	Financial shock (2001 Ansett collapse)	Executive level leaders (airport CEO, chairman, senior managers, n = 9)	Top management teams	Strategic decision making	Qualitative	The TMT responded to the crisis by centralizing authority and making quick strategic decisions based on less in-depth analysis, more intuition, limited consultation and by considering fewer strategic alternatives than they normally would.
Buyl et al. (2019)	2008 GFC	Executive level leaders (CEO, n = 92)	Corporate governance	Narcissism, risk taking	Quantitative (archival)	CEO narcissism positively influenced riskier bank policies pre-GFC, especially when risk-taking compensation policies were in place. This effect was dampened when board monitoring was more effective. Also, banks led by narcissistic CEOs pre-GFC experienced a slower recovery post-GFC.
Elliott and Stead (2018)	2008 GFC	Executive level leaders (women business leaders, n = unknown)	Gender and leadership	Women as: leaders and feminine; as credible and lacking credibility; as victims and as their own worst enemies.	Qualitative	The media constructions of women leaders during the GFC present a greater acceptability of femininity but not enough to challenge the association of masculinity with leadership and the male body.
Enander et al. (2009)	Severe storm (Gudrun)	Political leaders (municipal chief officers and politicians, n = 19)	Top management teams	Sensemaking and decision making	Qualitative	Leader decisions and reactions to the crisis were influenced by the complex interplay between: (1) the way they evaluated the crisis; (2) characteristics of the leaders and crisis, and (3) the social context within which the event occurred (i.e., following the Boxing Day tsunami).
Fralich and Sources of power: CEO-Chairman; ownership; prestige; expert	Quantitative (archival)	Powerful CEOs paid smaller premiums than weaker CEOs because they were better equipped to deal with the increased information uncertainty and asymmetry during the GFC, either through risk-averse behavior or better assessment of target quality.	Papadopoulos (2018)	2008 GFC	Executive level leaders (CEO, n = 242)	CEO power
Gupta et al. (2018)	Significant industry-wide downturn (shock)	Executive level leaders (CEO, n = 3,728)	Upper echelon theory	CEO power	Quantitative (archival)	Powerful CEOs ultimately undermine company performance during turbulent times.
Hede (2011)	Severe storm (Gudrun)	Political leaders (municipal chief officers and politicians, n = 13)	Top management teams	Leader learning; crisis preparedness	Qualitative	Lessons from the crisis are more complex than exposing strengths and weakness or creating 'to-do-lists', more importantly is the need to develop and understand mental models of what the crisis manager might face and have to deal with.

Table 3 (continued)

Author(s)	Crisis Description	Leader level (and Description)	Leadership or Related Theory	Leader Behavior or Style	Study Design	Key Findings
Hsu (2017)	2014 Kaohsiung underground gas explosion	Political leaders (village heads, n = 13)	Public leadership	Sensemaking and decision making	Qualitative	Village heads' personal attributes (actively or inactively seeking information and responding to the disaster), command authority, and knowledge played an important role in their decision making around the time of the explosion.
Jensen et al., (2021)	COVID-19 pandemic	Political leaders (US Governors, n = 50)	Charismatic leadership	Charismatic rhetoric	Mixed-method	A one standard deviation higher level of charisma may result in a roughly 6 percent lower level of COVID-19 deaths, cumulatively over a 10-week period. Charisma fuels beliefs in political conservatives, but not liberals. Sensemaking and public speaking after crises is seen as 'meaning making' by communicating the broader impacts of the crisis to citizens, media and other stakeholders.
Jong et al. (2016)	Downing of Malaysia Airlines Flight MH17	Political leaders (Dutch mayors, n = unknown)	None	Sensemaking	Qualitative	Assumptions and representations of leaders as 'masters of the universe' contributed to the demise of a mutual building society during the 2008 GFC and could potentially contribute to the next financial crisis.
Knights and McCabe (2015)	2008 GFC	Executive level leaders (CEO, n = 2)	Gender and leadership	CEO power	Qualitative	The media's ethical leadership depiction of two banking CEOs (through text and image) shifted with the changing context of the 2008 GFC.
Liu (2017)	2008 GFC	Executive level leaders (CEO, n = 2)	Ethical leadership	Sensemaking; leader aggression; competitive masculinity; paternalistic masculinity; humility and accessibility	Qualitative	This study highlights the role of the media in the active construction of leader's narratives and questions the notion that leaders have a true self and that their values, beliefs and attitudes will be accepted by others as 'authentic'.
Liu et al. (2017)	2008 GFC	Executive level leaders (CEO, n = 4)	Authentic leadership	Confidence; resilience; self-awareness & exposure; hope & optimism; moral character	Qualitative	This study found that crisis plans created prior to the COVID-19 pandemic were inadequate due to the long duration and highly uncertain nature of the crisis. Instead, higher education leaders applied guiding principles on the fly to support their decision making.
Liu et al. (2021)	COVID-19 pandemic	Leaders of mixed types (US higher education leaders, n = 37)	Crisis preparedness and responsiveness	Organizational values; crisis planning; ethics	Qualitative	Top management teams need to provide clear structure, role clarity, timely and decisive actions, foster ethical and moral principles, and be involved in business-critical negotiations.
Mora Cortez and Johnston (2020)	COVID-19 pandemic	Leaders of mixed types (US, Europe & Latin America companies, n = 31)	Top management teams; social exchange theory	Decision making; coordination; support; moral enhancement	Qualitative	School principals were successful in leading their schools through the disaster because dispositional, relational and situational factors influenced their leadership decisions and actions.
Mutch (2015)	2010/2011 Christchurch earthquakes	Executive level leaders (principal, n = 4)	Crisis leadership (Boin et al., 2010)	Sensemaking and decision making	Qualitative	Female-led banks hold about 5–6 % more equity capital than male-led banks, leading to a negative relationship between female CEO/ Chairs and bank default risk during the GFC.
Palvia, Vähämaa, and Vähämaa (2015)	2008 GFC	Executive level leaders (CEO, n = 6,729)	Gender and leadership	Risk taking	Quantitative (archival)	Findings demonstrated the influence of gendered leadership during the initial response to the pandemic, but the positive effect of gender representation on short-term outcomes was limited.
Park (2021)	COVID-19 pandemic	Political leaders (OECD country leaders, n = 35)	Gender and leadership	Democratic norms; women representation in decision making positions	Quantitative (archival)	Firms with narcissistic CEOs experienced performance declines at the start of the GFC, and performance gains post-GFC that differ significantly from those under less narcissistic CEOs.
Patel and Cooper (2014)	2008 GFC	Executive level leaders (CEO, n = 392)	Upper echelon theory	Approach-avoidance motivation; narcissism	Quantitative (archival)	

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Table 3 (continued)

Author(s)	Crisis Description	Leader level (and Description)	Leadership or Related Theory	Leader Behavior or Style	Study Design	Key Findings
Patelli and Pedrini (2014)	2008 GFC	Executive level leaders (CEO, n = 332)	Communicative action theory; impression management theory	Certainty; optimism; activity; realism; and commonality	Mixed-method	CEO letters of the best performing firms are the most optimistic (a proxy for impression management), while optimism was a significant predictor of future firm performance.
Prouska and Psychogios (2019)	Greek economic crisis (2012–2014)	Frontline/ middle level managers (small enterprise line managers, n = 28)	Leader voice	Sportsmanship silence; cooperative silence; cynical silence	Qualitative	Front line leaders facilitated employee silence due to their cynicism over the organization's continual changing of decisions which created worsening work conditions throughout the crisis.
Riaz et al. (2016)	2008 GFC	Executive level leaders (CEO, n = 11)	Upper echelon theory	Leader epistemic authority rhetoric	Qualitative	CEOs used four rhetorical strategies to strengthen their epistemic authority following the 2008 GFC, including: (1) rational guarantees, (2) normative responsibilities, (3) critiquing judgments, (4) questioning motives. Ratings of leadership ability from US CEO faces predicted their companies' financial performance pre-GFC, but not post-GFC. Power predicted German CEO success post-GFC, and US CEO success pre-GFC (but not post-GFC).
Rule and Tskhay (2014)	2008 GFC	Executive level leaders (CEO, n = 107 & n = 176)	Facial judgements and leadership	Leader power; warmth	Quantitative (experiment)	Greedy CEOs showed a reduced concern for stakeholder welfare, tending to forgo long-term investments that required short-term financial sacrifices. There was a negative relationship between CEO greed and CSR.
Sajko et al. (2021)	2008 GFC	Executive level leaders (CEO, n = 301)	Corporate Social Responsibility; organizational resilience	CEO Greed	Quantitative (archival)	US States with women versus men Governors reported fewer deaths from COVID-19 and expressed more empathy and exuded greater confidence.
Sergent and Stajkovic (2020)	COVID-19 pandemic	Political leaders (US Governors, n = 55)	Gender and leadership	Confidence; empathy	Mixed-method	This study highlights the role of public (gubernatorial) leadership in drawing a clear picture of needs, marshalling support from political partners, and convincing skeptics post hurricanes Floyd and Katrina.
Smith et al. (2018)	Hurricanes Floyd (1999) & Katrina (2005)	Political leaders (US Governors, n = 2)	Public leadership	Understanding local needs; immediate action to provide assistance; cooperative actions	Qualitative	Promotion-focused (reward) communication is more effective than prevention-focused (threat) communication in crisis situations.
Stam et al. (2018)	Financial shock	Leaders of mixed types (US Presidents, professors, & CEO)	Leader self-regulation	Promotion versus prevention focus	Mixed-method	Firm compensation actions were associated with a growth in positive sentiment over the initial months of the pandemic when made by CEOs with high benevolence, whereas firm safety actions led to growth in positive sentiment only when made by CEOs with low benevolence.
Steinbach et al. (2021)	COVID-19 pandemic	Executive level leaders (CEO, n = 59)	Attribution theory	Employee safety actions; compensation actions; CEO benevolence	Quantitative (archival)	The 2008 GFC was associated with an increase in directive leadership, which was stronger in the manufacturing sector and in countries with a high power distance.
Stoker et al. (2019)	2008 GFC	Frontline/ middle level managers (managers, n > 20,000)	Threat-rigidity hypothesis	Directive and participative leadership	Quantitative (archival)	Each leaders' belief in their ability to control events, self-confidence, and economic pressure partly explained their sensemaking, whereas cognitive complexity and openness to information did not.
Swinkels (2020)	Eurozone crisis	Political leaders (European Council leaders, n = 12)	Collective political leadership	Sensemaking; meaning making; belief changes (economic philosophy)	Qualitative	The relational activation of resilience model describes the formation of relational networks that are initiated when leaders signal that an organization is in crisis and are subsequently used by leaders for sensemaking and sensegiving.
Teo et al. (2017)	2003 SARS epidemic	Leaders of mixed types (healthcare leaders, n = 7)	Relational activation of resilience model	Sensemaking & meaning making	Qualitative	

Table 3 (continued)

Author(s)	Crisis Description	Leader level (and Description)	Leadership or Related Theory	Leader Behavior or Style	Study Design	Key Findings
Tourish and Hargie (2012)	2008 GFC	Executive level leaders (CEO, n = 4)	Metaphor; framing theory	Wisdom of the crowd; passive observers of events; victims of the crisis; penitent learners	Qualitative	The main intent of bankers was to frame public discussion of the GFC in terms which diminishes their responsibility for what happened, while the metaphors they employed have the potential to minimize learning from failure.
van Esch and Swinkels (2015)	Eurozone crisis	Political leaders (European political leaders, n = 6)	Leadership traits	Sensemaking; belief in the ability to control events; self-confidence; conceptual complexity; openness to information	Qualitative	A leader's belief that they can control events, their self-confidence, and economic pressure provide a partial explanation of how they make sense of a crisis.
Williams et al., (2021)	COVID-19 pandemic	Political leaders (US President Donald Trump and contender, Joe Biden)	Leadership traits; charismatic leadership	Trait honesty-humility; dark triad traits, attributed charisma	Quantitative (survey)	Voter perceptions of trait honesty-humility (dark triad traits) positively (negatively) influenced attributed charisma and negatively (positively) influenced leader rejection. COVID-19 crisis anxiety moderated the negative relationship between charisma and rejection for Trump, increasing the likelihood of rejection.
Windsor et al. (2014)	Natural disasters in China, Egypt & Cuba	Political leaders (Mao Zedong, Hosni Mubarak, Fidel Castro)	Autocratic leaders	Sensemaking; blaming or credit claiming language	Qualitative	Autocratic leaders use language strategically following natural disasters by blaming and credit claiming to maximize their time in office.
Zhang et al. (2012)	2008 China earthquake	Frontline/ middle level managers (hospital managers, n = 146)	Sensemaking theory; transformational leadership theory; leader-member exchange theory	Transformational leadership; emotional control; value congruence	Quantitative (survey)	Leader-follower value congruence mediates the relationship between transformational leadership and leadership effectiveness. Also, leader emotional control and quality of leader-member exchange moderates the relationship between transformational leadership and leader-follower value congruence.
Zhuravsky (2015)	2011 Christchurch earthquake	Leaders of mixed types (nurse/clinical managers, n = 4)	Shared leadership	Formal and informal leadership	Qualitative	Formal crisis-leadership themes included: decision making, ability to remain calm, and effective communication. Informal crisis-leadership themes included: motivation to lead, autonomy, emotional leadership, and crisis as opportunity.
b. Internal and unintentional crises						
Bauman (2011)	Product harm crisis (2007 Viracept recall)	Executive level leaders (corporate leaders, n = unknown)	Ethical leadership	Acknowledge; apologize; act	Qualitative	The author concludes that the ethical approach that most effectively addresses stakeholder reactions to unintentional harm is to acknowledge, apologize and act.
Cowen and		Montgomery (2020)	Product harm crises (OTC pharma products)	Executive level leaders (CEO)	Gender and leadership	Leader gender stereotypes (control vs agentic)
Quantitative		(experiment)	Unqualified apologies were more effective than other apologies (qualified apologies, sympathy) when delivered by a female CEO, but did not differ when delivered by a male CEO.			
Danielsson and Sjöstedt-Landén (2020)	School fire in Sweden	Leaders of mixed types (Principal, teachers, n = unknown)	Gender and leadership; implicit leadership theories	Masculine leadership behaviors	Qualitative	In this study, leader normativity worked to restore the strong male leader as the given center of the event, regardless of whether this matched the significance in the context of doing crisis management.

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Table 3 (continued)

Author(s)	Crisis Description	Leader level (and Description)	Leadership or Related Theory	Leader Behavior or Style	Study Design	Key Findings
Gorn et al. (2008)	Product harm crises (drug side effects)	Executive level leaders (CEO)	None	Babyfacedness (trait honesty)	Quantitative (experiment)	Examines the effects of babyfacedness on trustworthiness and perceptions of CEO, finding that trait inferences of honesty are made spontaneously but are corrected in more severe crises if cognitive resources are available.
Kashmiri and Brower (2016)	Product harm crisis (S&P 500)	Executive level leaders (Leaders of S&P 500 firms, n = 116)	Top management teams	Self-serving leadership	Quantitative (archival)	Self-serving managers under-invest in product quality-related systems and processes, resulting in low product quality levels. However, higher managerial ownership, higher family ownership, and the more influence marketing executives have in the TMT helps to align manager and shareholder incentives, resulting in a higher strategic focus on quality and lower likelihoods of a product harm.
Kashmiri et al. (2017)	Product harm crisis (general)	Executive level leaders (CEO, n = 395)	Upper echelon theory	Narcissism	Quantitative (archival)	Firms led by narcissistic CEOs exhibit a higher rate of new product introductions and a greater proportion of radical innovations in their new product portfolios but are also more likely to encounter product harm crises.
Laufer et al. (2018)	Product harm crises (laptop computers in South Korea & China)	Executive level leaders (CEO)	Power distance	CEO versus other spokespersons	Quantitative (experiment)	When the CEO was spokesperson during a product harm crisis in South Korea and China, consumers with high (versus low) levels of power distance had higher (versus lower) future purchase intentions. In South Korea, higher levels of power distance led to increased brand trust and increased future purchase intentions when the CEO was spokesperson.
Madera and Smith (2009)	Product harm crisis (OTC drug)	Executive level leaders (CEO)	Leader competence and legitimacy; leader emotions	Accept versus not accept responsibility; anger and sadness	Quantitative (experiment)	Leader sadness was evaluated more favorably than anger, and participants' emotion mediated the relationship between leaders' emotion and leader evaluation. Also, accepting (not accepting) responsibility for the crisis led to more (less) favorable evaluations.
Montgomery and Cowen (2020)	Product failure (defective motorcar fuel system sensor)	Executive level leaders (CEO)	Gender and leadership	Leader gender stereotypes (communal vs agentic)	Quantitative (experiment)	People respond more negatively to ethical failures for female versus male leaders. However, less negatively for female versus male leaders for competence failures.
Post et al. (2019)	Product failure (aeronautical company)	Frontline/middle level managers (production team leaders)	Gender and leadership; leader trust (evaluative & behavioral)	Interpersonal emotional management; relational behaviors	Quantitative (experiment)	A female leader's trust advantage during a crisis is attributed to the use of relational behaviors, but only when the crisis consequences are known.
Seeger and Ulmer (2002)	Industrial fire	Executive level leaders (CEO, n = 2)	Ethical leadership	Virtues of responsiveness; victim support; rebuilding & renewal	Qualitative	Both leaders displayed the virtues of responsiveness, victim support, rebuilding and renewal after the fires.
Sommer et al. (2016)	Chronic hospital bed shortage	Frontline/middle level managers (nurse unit managers, n = 52)	Full range leadership model	Transformational leadership; management-by-exception (MBE) passive; MBE active leadership	Quantitative (survey)	Transformational leadership was associated with higher positive and lower negative affect, which in turn predicted higher team member resilience. Opposite effects were found for MBE passive leadership.
Verčič et al. (2019)	Oil tanker environmental accident	Executive level leaders (CEO)	None	Apology or sympathy response to crisis	Quantitative (experiment)	Apology rather than sympathy results in higher post-crisis reputation of the organization and speaker, regardless of the source (i.e., CEO or spokesperson).
Zemba, Young, and Morris (2006)	Product harm crisis (toxic chemical spill)	Executive level leaders (CEO)	Collective agency orientation	Blame attribution (from perceivers)	Quantitative (experiment)	Perceivers who blame an accident on the organization as a collectivity are also likely to extend blame to a manager.

Table 3 (continued)

Author(s)	Crisis Description	Leader level (and Description)	Leadership or Related Theory	Leader Behavior or Style	Study Design	Key Findings
Zhang et al. (2021)	Product harm crisis (general)	Executive level leaders (CEO)	Ethical leadership; CEO turnover	Unethical and deceptive financial reporting	Quantitative (archival)	Income-increasing earnings management results in customer retention in the short-term and reduces the likelihood of CEO turnover. However, in the long term, there is a higher likelihood of financial restatement and worsening future performance.
c. Internal and intentional crises						
Graffin et al. (2013)	2009 British MP expense scandal	Political leaders (British members of Parliament, n = 644)	Ethical leadership; hazards of high status	Leader transgression; status and turnover	Quantitative (archival)	High-status MPs were not more likely to abuse the expense system than low status MPs but were more likely to be targeted by the media and voters for their transgressions.
Grover and Hasel (2015)	Sex scandals	Leaders of mixed types (political leaders, n = 9 & CEOs, n = 6)	Ethical leadership	Leader integrity; denial; apology; and atonement	Qualitative	Atonement from sex scandals is possible if the behavior was neither too extreme or out of character and the leader had sufficient power.
Hannah and Zatzick (2008)	Corporate scandals (e.g., Enron, WorldCom)	Executive level leaders (CEOs, senior executives)	Ethical leadership	Ethics; morals; and integrity	Qualitative	Consistent with an entrepreneurial worldview, landmark scandals had no impact on the portrayal of leaders in the US business press.
Klein and Eckhaus (2017)	Enron bankruptcy	Executive level leaders (Enron managers)	Top management teams	Sensemaking & sense giving	Qualitative	Employees could sense the impending crisis and undertook informal scanning as sensemaking of their unstable reality. At the same time, managers used sensegiving via more positive words in their emails. This tendency increased after the crisis broke as a form of cognitive dissonance between their identity and reality.
Van Scotter and Roglio (2020)	CEO misconduct	Executive level leaders (CEO, n = 259)	Leadership trait theory	Five factor model personality; narcissism; antisocial personality disorder	Quantitative (archival)	Strong relationships between CEO misconduct and personality profile similar to the prototypical profile of persons diagnosed with clinical personality disorders.
Varma (2020)	Forceful removal of airline passengers	Executive level leaders (CEOs of United & Delta Airlines)	Responsible leadership	Actions: (1) early disclosure & accept responsibility; (2) share information; (3) adopt governance leadership; (4) rebuild confidence; (5) restructure for credibility; (6) demonstrate social concern; (7) apologize	Mixed-method	United Airlines CEO followed most of the responsible leadership actions, which explained 70 % of the variation in share price. Whereas, only 50.6 % of the variation in Delta Airlines share price was explained by the more rational and dispassionate reaction from the CEO.
von Walter et al. (2016)	Employee corruption	Executive level leaders (High-ranking manager, management board)	None	Management board: corrective versus ceremonial response	Quantitative (experiment)	Frontline employees were more supportive after a scandal caused by a high-ranking manager versus low-ranking employee, when a corrective response was used, whereas the reverse was the case for a ceremonial response.
Zona et al. (2013)	CEO fraud (bank)	Executive level leaders (CEO)	(Un)ethical leadership, charisma (dark side)	Lack of moral values; high CEO aspirations; CEO charisma	Qualitative	A cycle of CEO charisma, misrepresentation, and excess of resources reinforces CEO hubris and overconfidence. This becomes unsustainable over time and creates a fragile organizational setting, exposing the firm to detection by judicial courts for illegal behavior leading to public awareness and corporate scandal.
d. External and intentional crises						
Birkeland et al. (2017)	2011 Oslo bombing	Frontline/ middle level managers (immediate superior)	Supportive leadership	Supportive; fair; and empowering	Quantitative (survey, natural experiment)	Employees with high posttraumatic stress viewed their immediate leader to be less supportive. However, overall employee perceptions of leadership were relatively stable pre- and post-crisis. Further, the control group (not exposed to the attack) had similar views of leadership as those exposed to the attack.

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Table 3 (continued)

Author(s)	Crisis Description	Leader level (and Description)	Leadership or Related Theory	Leader Behavior or Style	Study Design	Key Findings
Bligh et al. (2004)	9/11 attacks	Political leaders (US President George W. Bush)	Charismatic leadership	Charismatic rhetoric	Mixed-method	President Bush's rhetorical language, and his portrayal in the media, became more charismatic following 9/11.
Boehm et al. (2010)	2006 Lebanon War	Political leaders (grassroot community leaders)	Full range leadership model; community leadership	Transformational and transactional leadership	Quantitative (survey)	At times of crisis, community members expect grassroot leaders to display both transformational and transactional leadership.
Combe and Carrington (2015)	Changing Government Policy	Leaders of mixed types (CEO trustees, n = 2, regional leaders, n = 6)	Top management teams	Sensemaking; leader beliefs and values; cognitive consensus	Qualitative	Considerable individual differences in cognitive content were identified when starting to make sense of the crisis, while increasing cognitive consensus arose within well-defined teams over time based on similarities in objectives and cause-effect beliefs.
North et al. (2013)	9/11 attacks	Frontline/middle level managers	Supportive leadership	Supporting employees' return to work; workplace memorials and commemorative gatherings; personal sharing and support.	Qualitative	Managers felt ill-prepared to manage the magnitude and intensity of employees' emotional responses post 9/11. Employees attributed their emotional recovery to the actions of managers in their rapid return to work.
e. Multiple crisis types (External - intentional and unintentional)						
Davis and Gardner (2012)	9/11 attacks, Hurricane Katrina	Political leaders (US President George W. Bush)	Charismatic leadership	Leader charismatic rhetoric; leader effectiveness	Mixed-method	Overall, crisis serves as an antecedent for charismatic leadership as President Bush's rhetoric became more charismatic after the 9/11 attacks compared to pre-crisis levels. However, Bush's continued use of charismatic rhetoric may have become unproductive over time as each crisis passed.
Williams et al. (2009)	War, national deficit, terrorism, and foreign relations	Political leaders (US President George W. Bush and contender, John Kerry)	Charismatic leadership	Attributed charisma; value congruence; decisiveness	Quantitative (survey)	There was a negative (positive) relationship between crisis and charismatic leadership for George W. Bush (John Kerry). The relationship between crisis and voting behavior was mediated by charismatic leadership for both Bush and Kerry.
Williams et al. (2012)	2008 GFC, Iraq and Afghanistan wars	Political leaders (2008 US Presidential candidates, Obama versus McCain)	Authentic leadership; attributed charisma; value congruence	Authentic leadership; charisma perceptions	Quantitative (survey)	Crisis perceptions were positively related to attributed charisma but not perceptions of authentic leadership. Value congruence moderated the relationship between cynicism and attributed charisma for Obama (but not McCain), while attributed charisma had augmenting effects over authenticity in predicting voting behavior.

Since external-intentional crises have an identifiable "enemy" that blame can be attributed to, leaders in this quadrant appeared to increase their charismatic rhetoric and transformational leadership behaviors in response to a crisis, attempting to unify their followers by delivering an "us versus them" message (e.g., Bligh et al., 2004; Davis & Gardner, 2012). By creating this connection with followers, leaders provided a source of comfort for them and offered hope that the common enemy would be made to atone for their wrongdoing. In turn, followers respond favorably to leaders who communicated charismatically and demonstrated transformational leadership in times of crises compared with periods of normality (Bligh et al., 2004; Boehm et al., 2010). To sum up the findings in this quadrant, when facing an external-intentional crisis effective crisis leaders provide followers with comfort, support, and hope for the future.

Internal crises

(3) Leaders as spokesperson: Apologizing after internal-unintentional crises. The second largest quadrant in our review was internal-unintentional crises ($k = 15$, 21.1 %), with most articles focused

on product defects and failures ($k = 11$, 73.3 %). Product harm crises pose significant threats to organizational reputation and survival, typically resulting in plummeting sales and punitive action by regulators. Such crises are problematic for organizational leaders, as people often attribute intentionality to those who have "allowed" harm to occur, even though it was unintentional (Ames & Fiske, 2013; Dasborough & Ashkanasy, 2002). In addition, organizational leaders are often held to a higher standard of foresight and integrity than employees, regardless of the type of harm caused (Dasborough & Harvey, 2017). Hence, product harm crises can significantly damage organizational reputation if leaders fail to effectively manage stakeholder perceptions (Bauman, 2011), and returning stakeholder trust to the level held prior to such crises is difficult (Madera & Smith, 2009).

The overarching theme in response to internal crises that were unintentional (but may nonetheless be attributed to the organization), concerns the importance of how leaders communicate with external stakeholders about a crisis, which casts the *leader as spokesperson*. In this regard, studies showed that leaders tried to manage how a crisis within their organization was perceived by external stakeholders through impression management tactics (Verčič, Verčič, & Coombs,

2019). To repair the reputational harm that an internal-unintentional crisis may cause, leaders communicated in ways that enhanced trust, and were careful with both the content and authenticity of their communications (Coldwell, Joosub, & Papageorgiou, 2012). When leaders communicated honestly, transparently, and with integrity, they appeared to be more effective at minimizing reputational damage (e.g., Bundy et al., 2017; Pearson & Clair, 1998). The studies in this quadrant found that trust could be repaired if leaders took responsibility for a crisis by publicly apologizing, disclosing failures, accepting accountability, and providing compensation where appropriate (e.g., Coldwell et al., 2012; Gillespie & Dietz, 2009; Madera & Smith, 2009). Conversely, when they communicated in ways that external stakeholders deemed inappropriate, they risked increasing the fallout caused by the crisis (e.g., Madera & Smith, 2009).

Another element of the *leader as spokesperson* theme concerns who speaks about the crisis on behalf of the organization. When it came to organizational trust repair, it is not just what was said, but who was saying it. As reported previously, most of the studies in this quadrant focused on the CEO, as CEO's are often the spokesperson following internal-unintentional crises. However, the choice of spokesperson for this type of crisis may be less important than the communication strategy. For instance, Verčič et al. (2019) found that apology was an effective strategy in protecting organizational reputation, regardless of whether it was the CEO or another spokesperson who delivered the message.

Finally, several studies in this quadrant found that leader gender can have an impact on stakeholder responses. For example, Cowen and Montgomery (2020) found that apology type made no difference to future purchase intentions when delivered by a male CEO, whereas an unqualified versus qualified apology from a female CEO led to an increase in future purchase intentions. Another study by the same coauthors found that while people respond more negatively to ethical failures by female (versus male) leaders, they respond less negatively for competence failures by female leaders (Montgomery & Cowen, 2020). The evidence from studies in this quadrant suggest that external stakeholders might be more trusting of female leaders who apologize unreservedly in response to internal-unintentional crises, and this may be due to a female leader's superior use of relational skills in contrast to male leaders (see Post et al., 2019; Ryan, Haslam, Hersby, & Bongiorno, 2011). To conclude, in the case of internal-unintentional crises, the choice of spokesperson (CEO or other spokesperson, male/female), communication content and tactics, and leader attributes matters for trust and reputation repair.

(4) Leaders as sinners: Atoning for internal-intentional crises. The eight articles in the internal-intentional crises quadrant (11.59 %) mostly examined transgressions committed by strategic level leaders (CEO's and politicians). A variety of scandals were examined in this quadrant, including the inappropriate use of public funds (Graffin, Bundy, Porac, Wade, & Quinn, 2013), bribery (von Walter, Wentzel, & Tomczak, 2016), fraud (Zona, Minoja, & Coda, 2013), and sexual transgressions (Grover & Hasel, 2015). This quadrant is somewhat unique compared to the other three quadrants as it focuses on unethical (and in some cases illegal) behaviors committed by the leaders themselves. These behaviors are typically viewed by both internal and external stakeholders as immoral (e.g., Graffin et al., 2013). Anti-social personality traits such as narcissism (Van Scotter & Roglio, 2020), as well as arrogance, dishonesty and immorality (Zona et al., 2013) featured among many of the leaders in these studies. Hence, the theme of *leaders as sinners*.

Half of the studies in this quadrant examined how leaders *atoned for their sins* (i.e., mitigated the damage caused by the internal-intentional crisis that they caused, Klein & Eckhaus, 2017; Grover & Hasel, 2015; Hannah & Zatzick, 2008; Varma, 2020). Like the previous internal-unintentional crises quadrant, leaders were viewed more positively if they publicly apologized, were accountable, and made promises to reform (e.g., by promoting ethical conduct, Grover & Hasel, 2015;

Varma, 2020). However, in contrast to internal-unintentional crises, where leaders' trust repair actions were typically directed toward external stakeholders, leader responses to internal-intentional crises also targeted internal stakeholders. This is because internal stakeholders (i.e., employees) were mostly uninvolved in the scandal but were nonetheless stigmatized by association (Warren, 2007). Leaders who did not respond to transgressions adequately faced the risk that employees would withhold their support (e.g., expressing a lack of trust in leadership) or undermine the organization (e.g., leaking information to the media). For example, when senior leaders offer ineffective or ceremonial crisis responses (von Walter et al., 2016).

Studies in this quadrant also examined factors that may affect stakeholder perceptions of leaders who *atone for their sins*. Although high status leaders were more likely to be criticized for their transgressions compared to low status leaders, they were also in a better position to survive the crisis they caused than those with less power (Graffin et al., 2013). The evidence from this quadrant indicates that when an appropriate corrective response is taken by a leader who has transgressed, atonement is possible if the leader has sufficient power and is highly ranked in the organization (Grover & Hasel, 2015; von Walter et al., 2016). Like the prior quadrant about internal-unintentional crises, rebuilding trust and reputational repair was critical for leaders who retained their position. However, studies in this quadrant indicated that internal and external stakeholder forgiveness may only be successful for some leaders (those with power) and in the context of certain transgressions. Some internal-intentional crises are perceived to be too unethical to recover from, as they involve illegal leader behavior (i.e., rape of an underage female; Grover & Hasel, 2015; fraud, Zona et al., 2013).

Similarities and differences across crisis types

From our analysis of the articles in our review, we identified themes that were unique to each type of crisis. In the case of external-unintentional crises, effective crisis leaders used their power and charisma to shepherd their people to safety throughout all stages of the crisis. For crises that are external and intentional, the saint-like leader supported and empowered their followers. When the crisis is internal, but unintentional, the successful crisis leader was an honest spokesperson who sought to repair trust and uphold the organization's reputation. When the crisis is internal and intentional, effective crisis leaders atoned for their or their organizations' sins. While these themes were unique to each quadrant, there were also overlapping themes.

Based on the review findings, there appears to be two recognizable leadership themes distinguished by a demarcation between internal and external crisis types. Internal crises arise from within the organization, and regardless of whether the crisis is intentional (e.g., sex scandal) or unintentional (e.g., product defect), most scholars appeared to approach crisis leadership from a trust repair (Gillespie & Dietz, 2009) and/or ethical leadership (Brown & Trevino, 2006) perspective. It also appears that the focus was largely on external stakeholders and ways that the leader could regain the trust of the public after a crisis from within their organization.

In contrast, for external crises that arise from outside the organization, scholars have tended to focus on examples of charismatic-transformational leadership (Bass, 1990), regardless of whether the crisis occurred through intentional (e.g., terrorism) or unintentional means (e.g., natural disasters). In this case, the successful crisis leader used his or her charisma and individualized consideration to help boost follower optimism and to aid their emotional recovery after the crisis. In contrast to internal crises, where the focus of crisis leaders is mostly on managing external stakeholder perceptions, the focus during external crises was predominantly on protecting the lives and livelihood of internal stakeholders.

Directions for future research on crisis leadership

In this review, we utilized the crisis typology proposed by Coombs and Holladay (1996) to examine leadership within and across different crisis types. This approach allows us to provide evidence-based guidelines for future researchers to answer the call by Bavić et al. (2021) to study leadership across different crisis contexts. The findings from our thematic analysis suggest that leaders appear to respond differently to various crises in order to meet the needs of various stakeholders. Next, we outline several promising directions for future research based on the findings of our review.

Crisis leadership Antecedents and leadership across levels

A relatively narrow range of individual differences, mostly related to ethical leadership (Brown & Treviño, 2006), were found in the studies examined in this review. For example, narcissism and risk-taking (e.g., Buyl et al., 2019; Kashmiri, Nicol, & Arora, 2017; Patel & Cooper, 2014), greed (Sajko, Boone, & Buyl, 2021), optimism (e.g., Patelli & Pedrini, 2014), honesty (Gorn, Jiang, & Johar, 2008), and integrity (e.g., Grover & Hasel, 2015; Hannah & Zatzick, 2008). Hence, a promising area for future research would be to examine a broader range of leader individual differences (e.g., personality, cognitive abilities), and other characteristics (e.g., attitudes, abilities), as antecedents of crisis leadership and related outcomes. This approach is consistent with calls to examine meaningful combinations of leader traits and characteristics in explaining leadership (e.g., Zaccaro, 2012), as well as calls to examine leadership across contexts (Oc, 2018).

Also, given that we found different leadership themes for each type of crisis, another opportunity might be to identify individual differences and characteristics that relate to the crisis leadership themes that we identified from our analysis (i.e., leaders as shepherds, saints, spokespersons, and sinners). For example, examining the influence of leader self-regulation could enhance our understanding of how leaders make decisions under pressure as shepherds (e.g., Taylor-Bianco & Schermerhorn, 2006), and their unethical behavior as sinners (Joosten, van Dijke, Van Hiel, & De Cremer, 2014). In Table 4, we present several questions that could potentially guide future research examining the antecedents of each crisis leadership theme, as well as other questions that examine the multi-level nature of crisis leadership (which we discuss next and in the following methodological critique section).

In addition to the four crisis leadership themes arising from our review, we also identified several future research opportunities that are common across these themes. Specifically, we next offer suggestions to advance the study of crisis leadership by examining frontline leadership, gender and leadership, leaders as emotion managers, and stakeholder attributions of stability and intentionality. We believe these topics are highly relevant to advancing our understanding of crisis leadership, yet they have attracted relatively little interest from crisis leadership scholars to date.

Frontline crisis leadership

In our review, we found that most of the empirical studies were focused on strategic level leaders (e.g., CEOs). While we acknowledge the importance of the CEO as organizational spokesperson and leader of the crisis response, we also argue that frontline leaders are critical as they are more “hands on” than CEO’s (Dasborough & Scandura, 2021). Antonakis and Atwater (2002) would categorize the strategic leaders in our review as “Manor house” or “distal” leaders and suggest that the outcomes of this type of leadership is likely to be more prominent at the group level (e.g., team affective climate, safety climate), rather than at the individual level of analysis (e.g., individual performance, individual wellbeing).

Research on leader distance suggests that the effects of leader behavior depend on the level of analysis at which leadership is examined, and that these effects vary as a function of leader–follower distance (Antonakis & Atwater, 2002). Given that few studies in our review examined “close leaders” (i.e., frontline leaders), it would be interesting to investigate the dyadic relationships between frontline leaders and their employees. Leader-member exchange (LMX) theory could be used to examine the relationship between these leader-member dyads, as well as their shared emotional experiences of the crisis (Cropanzano, Dasborough, & Weiss, 2017).

Moreover, leader distance has already been examined with respect to charisma (Shamir, 1995), which according to our review makes it a promising avenue for leaders dealing with a crisis. Shamir (1995) suggests that charisma is fundamentally different for close (frontline) versus distal leaders (CEO’s). While CEO’s, as distal leaders, might be seen as charismatic due to the way they communicate their vision for recovery after a crisis, the charisma of frontline leaders will likely be associated with their observable behaviors and personal example as perceived by proximal followers during the crisis (see Ehrlich, Meindl, & Viellieu, 1990). Shamir (1995) found that close leaders are attributed charisma when they are sociable, considerate, humorous, and have expertise. In our review, while CEO consideration has been examined, sociability, humor, and expertise have been largely ignored, and these individual traits have not been examined in frontline leaders. Future research could investigate these leader traits in frontline leaders.

Gender and crisis leadership

Female leaders have been hailed as shining examples of crisis leadership during the COVID-19 pandemic (Garikipati & Kambhampati, 2021). Female global leaders have responded to the pandemic in ways that prior research and gender stereotypes would have predicted (i.e., as more truthful, decisive, and empathetic than their male counterparts, Chamorro-Premuzic & Wittenberg-Cox, 2020; Gipson, Pfaff, Mendelsohn, Catenacci, & Burke, 2017). However, the majority of studies we reviewed focused on male leadership during crises, despite women comprising 47.0 % of the US workforce. This underrepresentation of female crisis leadership research aligns with women being disproportionately under-represented in upper echelons of organizations, with women holding only 25.0 % of senior-level positions, 19.0 % of board seats, and 4.5 % of CEO positions (Offermann, Thomas, Lanzo, & Smith, 2020). Of the handful of studies investigating female crisis leaders, the focus has been on “the glass cliff” phenomenon (see Bruckmüller & Branscombe, 2010; Kulich, Lorenzi-Cioldi, Iacoviello, Faniko, & Ryan, 2015; Ryan & Haslam, 2005) rather than female leaders’ behaviors or styles. Future research should examine why women are seen to be more effective crisis leaders than men, what leadership behaviors they use, and which behaviors are learnable for all leaders, regardless of gender.

We call for not only studies of women leading in times of crisis, but also studies into feminine traits. Future research would benefit from a better understanding of feminine traits (e.g., sensitivity, compassion, warmth) and masculine traits (e.g., dominance, assertiveness, aggressiveness) during crises. While it may be assumed that feminine traits are most suited for crisis leadership (see Ryan et al., 2011), this may not always be the case. Different traits may be beneficial in response to different types of crises. For example, masculine rather than feminine traits may be associated with more effective crisis leadership in response to an external-intentional crisis (e.g., terrorist attack).

Crisis leaders as emotion managers

Noticeably absent in our review are studies of leader and follower emotions during a crisis, despite previous calls to examine this phenomenon (James et al., 2011). Facing a crisis at work can be stressful

Table 4

Future research directions.

Crisis Leadership Themes	Potential Research Questions
<i>Internal-Unintentional: Leaders as Shepherds</i> • Leaders guide and protect their followers and organization by sensemaking and rapid decision making • Leaders use their power and charisma to guide and protect their followers and organization	Crisis Leadership Antecedents - What other sources of power do leaders employ to shepherd their organizations? Studies have examined referent power (charisma), but what about expertise? - Is charisma used differently by frontline leaders versus strategic leaders? Is it perceived differently depending on leader level? - What is the impact of leaders with a long-term orientation versus a short-term orientation during this type of crisis? - How does leader sensemaking differ for external versus internal stakeholders? Crisis Leadership Across Levels - How is crisis leadership shared by top management teams? - How do individuals respond to shared leadership during crises? - How do leaders use charisma at different time points? Is charisma more important at different stages of the crisis lifecycle?
<i>External-Intentional: Leaders as Saints</i> • Leaders comfort and support their followers, as they are victims who may still be fearful of the external threat • Leaders empower followers to help them return to a sense of normalcy	Crisis Leadership Antecedents - What traits help leaders provide comfort to their followers? For example, traits like emotional intelligence, empathic concern, or perspective taking? - How does leader self-regulation of their emotions during a crisis impact stakeholder responses to their actions? - How do leaders cope with the stress caused by this type of crisis? Crisis Leadership Across Levels - What role does organizational culture play in leader responses to crises? Are we more likely to see leaders as saints in organizations with compassionate cultures? - What role does organizational culture play in responsiveness to leadership through this type of crisis? - How does empowerment trickle down from strategic level leaders to frontline leaders?
<i>Internal-Unintentional: Leaders as Spokesperson</i> • Leaders can repair trust by communicating openly and honestly with their stakeholders about the crisis • Leaders with superior relational skills will be more trusted by stakeholders	Crisis Leadership Antecedents - What communication skills are required to respond to this type of crisis? - For crisis communication, when are feminine traits most appropriate? When are masculine traits most appropriate? - What communication training should leaders undertake to prepare for possible crises? Crisis Leadership Across Levels - What role does organizational climate of psychological safety have in leader honesty about organizational failures? - Is this type of crisis unique to large (public) firms? - How important is the timing of leader apologies in response to this type of crisis?
<i>Internal-Intentional: Leaders as Sinners</i> • Leaders who have done wrong seek forgiveness by apologizing • Leader status and power will determine if they can atone for their sins	Crisis Leadership Antecedents - Would the use of political skills help a leader recover from this type of crisis? - How does prior LMX relationship quality impact employee responses to leaders who have transgressed? - If stakeholders identify with an organization, does it mitigate or exacerbate the harm caused by this type of crisis? - Should leader response to these types of crises differ according to organization size? Crisis Leadership Across Levels - Is this type of crisis becoming more or less prevalent over time? - Can leaders fully recover over time from an internal-intentional crisis that they have caused? - How long does it take to recover the previous level of trust with other leaders, employees, and external stakeholders?

(Antoniou & Cooper, 2017; Halkos & Bousinakis, 2017), and even in non-crisis situations individuals often look to their leaders to alleviate stress and manage negative emotions at work (Niven, 2017). Silard and Dasborough (2021) suggest that effective leaders act as “emotion shields” for their followers by protecting them from high arousal negative emotions. We extend this notion to crisis situations, namely that leaders need the awareness and ability to manage the high arousal negative emotions likely to be experienced by their followers during a crisis.

To this end, leaders are expected to intervene proactively and effectively to “calm troubled waters” when followers experience strong negative emotions (Ashkanasy & Daus, 2002; Cropanzano et al., 2017). In addition, the stress arising from a crisis is likely to lead to negative emotions, not only for subordinates, but also for leaders themselves (Harms, Credé, Tynan, Leon, & Jeung, 2017). The external crises identified in our review include industry-wide financial shocks, natural disasters, terrorist attacks and pandemics. Crisis leaders themselves are likely to experience shock and/or trauma in response to these events, yet their followers may still expect them to be decisive, supportive, and in control of their own emotions. However, at present, there are no studies that examine leaders’ experiences as victims of crises themselves.

A leader’s ability to self-regulate their negative emotions will affect their capacity to process information and make sound and timely deci-

sions under pressure (Collins, Ersing, & Polen, 2017; Dasborough et al., 2021; Mumford, Friedrich, Caughron, & Byrne, 2007). This is important in times of crisis, and will likely influence the leader’s communication, behaviors, and stakeholder perceptions of them during a crisis (see also Collins & Jackson, 2015). Despite some insights into the role that leaders play in managing the emotions of others in crisis situations, our review sheds little light on how managers self-regulate their own emotions during a crisis. In fact, only three of the studies we reviewed (Liu et al., 2017; Madera & Smith, 2009; Sargent & Stajkovic, 2020) described any type of leader emotional response during crisis events. This is an important gap in the literature, which has been raised in several articles (e.g., Enander et al., 2009; Mutch, 2015). These gaps in our knowledge of leader emotional responses to crises offer unique research opportunities for scholars to pursue (see Table 4).

Stakeholder attributions of stability and intentionality

Our review uses Coombs and Holladay (1996) crisis typology that draws on attribution theory (Heider, 1958; Weiner, 1986) to classify crisis events. This typology explicitly includes two causal attributional dimensions: “locus” (internal or external) and “controllability” (intentional or unintentional). Although Coombs and Holladay (1996) mention the third dimension of “stability”, they do not explicitly map it to

their typology. However, stability is an important consideration that shapes attributions and responses to observed behaviors. In future crisis leadership research, it would be interesting to examine stability in relation to the causal attributions that stakeholders make about crisis leaders. Consider an organization with a history of product failures (internal-unintentional crisis), or a senior leader with a long-standing reputation for sexual misconduct (internal-intentional crisis). Stakeholders in both scenarios might be quick to attribute blame to the organization or senior leader for a similar crisis, even when the cause of the crisis was unintentional.

Attributions of intentionality are also a promising avenue for future research. Although the typology uses the label “intentional or unintentional”, it does not really tap into specific intentions or motives. Building on this typology by adding crisis leadership, we could examine attributions of leader intentions in each quadrant. In the leadership literature, there have been studies of various motives, such as performance promotion motives and injury initiation motives (Liu, Liao, & Loi, 2012), as well as sincere intentions and manipulative intentions (Dasborough & Ashkanasy, 2004; Lin, Huang, Chen, & Huang, 2017). Future research could examine different types of attributions of intent that stakeholders make about leaders’ responses to various types of crises. For example, if a leader apologizes for committing fraud, is the apology perceived as sincere or manipulative? If a leader asks employees to work overtime during the COVID-19 pandemic, is this perceived as being driven by performance promotion motives or injury initiation motives? Such questions would help us understand how stakeholders respond to crisis leadership actions.

Practical implications for crisis leaders

The four themes that emerged from our review of the crisis leadership literature provide unique insights into the way leaders responded to various types of crises. Although this review was descriptive, rather than prescriptive, our findings suggest that certain leader responses are seen to be effective by various stakeholders during different crisis types. While the guidance we offer here may be relevant for all crisis types, each type of crisis also has unique properties and challenges (James et al., 2011). As such, we offer suggestions for crisis leaders according to the type of crisis they might face.

In the case of external-unintentional crises, leaders as *shepherds* are responsible for protecting and guiding various stakeholders. As such, they have an influential role in shaping how people perceive the impact of the crisis, and how stakeholders judge their response to the crisis. Indeed, how a leader responds to a crisis is crucial in determining its eventual fallout, since crises are perception-based (Bundy et al., 2017). External-unintentional crises, such as the COVID-19 pandemic, cause great uncertainty and leaders need to be cognizant of the major role they play in determining how stakeholders make sense of what is happening at each stage of the crisis life cycle (Weick, 1988). The way leaders respond to a crisis (i.e., what they say, how they say it, what actions they take, and how quickly they take them) influences stakeholders in at least three ways. First, it signals the urgency of the crisis at hand when leaders communicate without delay (Pearson & Clair, 1998). Second, it influences evaluations of the organization and its leadership when leaders communicate openly, honestly and clearly (Boin & Hart, 2003; Boin et al., 2005). Finally, it indicates who is responsible for mitigating the impact of the crisis when leaders ensure that everyone knows what they need to do (Bundy et al., 2017). This approach to crisis communication increases leader endorsement and follower support (Stam, van Knippenberg, Wisse, & Nederveen Pieterse, 2018).

During an external-intentional crisis, leaders need to monitor and manage a wide range of negative emotions that followers may potentially experience (e.g., shock, anxiety, fear, anger, fury, disgust, despair, guilt, and sadness, Birkeland et al., 2017). In response to such

high arousal negative emotions, leaders as *saints* need to provide comfort and support to others. To effectively lead their organizations through this type of crisis, leaders need a range of emotional skills to help manage the emotions of their people (e.g., by showing sympathy and compassion). Specifically, leaders need skills in emotion perception, emotion understanding, emotion regulation, and empathy (Antonakis, Ashkanasy, & Dasborough, 2009; Dasborough et al., 2021). Leaders also need to regulate their own emotions in response to a crisis (e.g., Dasborough & Scandura, 2021; Restubog, Ocampo, & Wang, 2020). For example, initial evidence suggests that leaders who express disappointment or sadness for the harm caused by a crisis are evaluated more favorably than those who express anger (Madera & Smith, 2009).

Beyond influencing the emotions of their followers and themselves, organizational leaders also need to provide hope and inspiration to a broader stakeholder group. To achieve this, leaders should display charisma to inspire recovery via expressing positive emotions (e.g., Bligh et al., 2004). Evidence demonstrates that charismatic leadership tactics can be taught (Antonakis, Fenley, & Liechti, 2011), so leaders should attempt to influence their followers using charismatic tactics in response to crisis events. This includes using verbal charismatic tactics (e.g., metaphors, moral conviction, sentiments of the collective, confidence, contrasts, lists, rhetorical questions) and non-verbal charismatic tactics (e.g., body gestures, animated voice tone, facial expressions). Leaders who actively employ these charismatic tactics can have a positive effect on others’ emotional responses during times of crisis.

In the case of internal crises generally, an organizational leader is seen as the one who takes responsibility for “making it right” with their stakeholders. To do this, they may need to apologize for any intentional wrongdoing committed by their employees or themselves, or they may have to express genuine regret for failures, even if they were unintentional (e.g., product failures, industrial accidents). It is possible for leaders to rebuild trust and repair relationships if leaders apologize and make amends (e.g., Varma, 2020). As such, it may be helpful to train leaders in Coldwell et al. (2012) “Responsibility Compass”¹⁰ as a way to respond to internal crises. However, the effectiveness of trust repair will often depend on the type of crisis that the leader is apologizing for.

For an internal-unintentional crisis, where the cause is seen as a competence violation, it may be effective for the leader as *spokesperson* to apologize. In contrast, this may be less effective for an internal-intentional crisis, where the cause is seen as an integrity violation by a leader (Janowicz-Panjaitan & Krishnan, 2009). Apologies for integrity violations are less likely to be accepted, especially if the leader has regularly displayed unethical behavior or misconduct. Often in the case of an internal-intentional crisis caused by a leader, the leader as *sinner* is quickly replaced (e.g., fraud, sexual misconduct) with a more ethical leader. To avoid this scenario, organizations need to select and promote leaders who can abstain from immoral or unethical behaviors (Brown & Treviño, 2006). Organizational leaders should also promote speaking up when individuals first become aware of ethical transgressions (e.g., Lucas & Fyke, 2014).

Finally, our review suggests that there is no “one-size-fits-all” crisis leadership style that is effective across all crisis types. For example, there is preliminary data attesting to the efficacy of leader flexibility during natural disasters (Teo, Lee, & Lim, 2017; Smith et al., 2018), yet there is also evidence that directive leadership is effective during economic crises (Stoker, Garretsen, & Soudis, 2019). In effect, there is insufficient and inconsistent evidence at this stage to establish a crisis leadership approach that is generalizable across all crisis types. However, it appears safe to say that seeking a “one-size-fits-all” solu-

¹⁰ This response involves: (1) early disclosure and acceptance of responsibility, (2) sharing information openly with the media, (3) adopting governance leadership to handle the event, (4) rebuilding confidence, (5) restructuring for credibility, (6) demonstrating social concern, and (7) apologizing publicly (see *Varma, 2020).

Table 5
Critique of crisis leadership research.

Methodological issue	Description	Future research recommendations
Qualitative studies		
Lack of acknowledgement and/or management of researcher-subject power imbalances	The vast majority of qualitative crisis leadership studies included in the review failed to report the presence or absence of power imbalance between the researcher(s) and participants.	Acknowledge and report the nature of power imbalance between the researcher(s) and participants, and how this imbalance will be handled.
Insufficient reporting of unexpected opportunities, challenges, and other events that occurred during the study	Almost all of the qualitative crisis leadership studies included in the review failed to report unexpected opportunities, challenges, and other events that occurred, and how these were handled.	Mindfully document and report unexpected opportunities, challenges, and events that occurred during the study and how these were handled (e.g., attrition, new participants, new data sources).
Deficient explanation of researcher positioning along the insider–outsider continuum	Most of the qualitative crisis leadership studies reviewed failed to explain the position of the researcher along the insider–outsider continuum (e.g., if an existing relationship existed with participants, rapport building).	Provide details in relation to the position of the researcher within the insider–outsider continuum, as well as strategies to ensure objectivity in the research process.
Unclear, or no mention of when theoretical saturation points were reached	Almost all reviewed qualitative crisis leadership articles failed to clearly identify the theoretical saturation point and/or describe the criteria or basis for judgment calls on ceasing data collection.	Describe in detail how theoretical saturation was reached and provide the criteria used to judge that the saturation point has been reached.
Lack of specificity of how qualitative data was analyzed	Many qualitative studies reviewed failed offer a description of their coding strategies (including first-order and second-order or higher-order codes) and how data were analyzed to reach higher levels of abstraction.	Clearly describe the type of coding strategy employed, including all stages of coding (e.g., open, axial, theoretical).
Insufficient reporting of inter-coder reliability	Most studies failed to report inter-coder reliability, which raises concerns over the reliability and validity of the findings.	Use more than one coder and calculate the inter-coder reliability for the coding. Then report the reliability when sharing the findings with others.
Lack of counterfactual evidence to verify the findings	Almost all the studies failed to consider alternative explanations to verify the findings.	Use thought experiments (i.e., imagined) to verify the findings, in order to enable the possibility of alternative explanations and new theoretical insights.
Quantitative studies		
Archival studies: Lack of external validity	Most datasets were derived from large, publicly traded companies in Western countries, which limits external validity of study findings.	Acknowledge generalizability limitations of study findings; actively search archival data sources that offer information from small and/or private firms and developing countries.
Archival studies: Overly simplified and/or arbitrary operational definitions of key variables	Key variables are operationalized in overly simplified ways using dummy and/or proxy variables. There was a tendency to use arbitrary cut-offs with no clear justification.	Offer clear justification for operational definitions of key variables and willingly acknowledge limitations of the dataset. Avoid using arbitrary cut-offs without justification.
Archival studies: Reverse causality and omitted variables	Most archival studies included supplementary analyses to test for reverse causality, and they included many control variables. However, the controls were limited to what was available in the database.	Design natural experiments for studying external crises. Use quasi-experimental procedures such as regression discontinuity in time (Hausman & Rapson, 2018) or diff-in-diff (to compare organizations/leaders who are or are not affected by the crisis, see Sieweke & Santoni, 2020). Use cross-lagged panel models with fixed effects to address reverse causality bias (Leszczensky & Wolbring, 2019). In the presence of unmeasured/unidentified confounds in the dataset, instrument variables that rely on an exogenous source of variance can be used to estimate effects consistently (Antonakis et al., 2010). Correlated random effects models can be used to address omitted fixed effects issue and to still include level 2 variables (Mundlak, 1978; Antonakis et al., 2010).
Cross-sectional survey studies: Omitted variables and lack of justification for control variables	Demographic variables (e.g., gender, age) were typically used as control variables without theoretical/empirical justification, increasing the potential for endogeneity.	The only way to overcome endogeneity concerns in cross-sectional studies is to use instrumental variable estimation, where the instrumental variables rely on exogenous sources of variance. Provide theoretical and/or empirical justification for the choice of control variables and explain how these relate to the dependent variable.
Cross-sectional survey studies: Common method variance issues	There was weak acknowledgement of common method variance issues and how these were addressed in most studies.	Design studies with keen attention to minimizing common method variance (e.g. collect multi-source data and longitudinal data); acknowledge and report the nature of common method variance issues and how these are addressed statistically and/or methodologically.
Cross-sectional survey studies: Potential measurement error	No modelling of measurement error or testing of cross-cultural measurement equivalence. Most studies only reported scale reliabilities, raising concerns about internal validity.	Incorporate and model measurement error in survey study designs.
Experimental: Unrealistic conditions	Most experiments used written vignettes. These are unrealistic in the context of a crisis. Only 1 study utilized a video; no study measured behavioral outcomes. This raises concerns about external validity.	Design more realistic experimental manipulations using immersive technology (e.g., virtual reality, videos); seek to measure behavioral outcomes.
Experimental: No behavioral outcomes	The experiments did not have any real behavioral outcomes with real consequences for participants.	Instead of asking about intended behaviors, test actual behaviors and design the outcomes to involve some cost for the participant so that it has meaning. This may also be achieved by using multiple study designs in one article.
Experimental: Unclear basis for sample size	Very few studies conducted a priori power analysis to determine appropriate sample size, decreasing the likelihood of detecting true effects.	Conduct a priori power analysis to justify sample size.

(continued on next page)

Table 5 (continued)

Methodological issue	Description	Future research recommendations
Experimental studies: Lack of covariates and instrumental variable procedures	By not controlling for possible relevant confounds, the treatment effect may not be as accurate. The use of instrumental variables strengthens the argument for causality.	Include covariates to increase the precision of the estimated treatment effect. This is particularly important to reduce bias if groups are not perfectly balanced. Use an instrumental-variable procedure (see Sajons, 2020) to increase confidence in causal findings.

tion is unlikely to be fruitful as each specific crisis type seems to require a different crisis leadership response.

A methodological critique of the crisis leadership literature

While the first aim of our review was to examine how different crises require different leader responses, the second aim was to critique the different methodologies that have been used in the literature. As such, guided by methodological best practice (e.g., [Aguinis & Solarino, 2019](#); [Antonakis, 2017](#); [Hill et al., 2021](#)), we also undertook a critical assessment of the quality and rigor of all qualitative, quantitative, and mixed methods studies included in this review. Even though most articles were published in reputable outlets, we identified several methodological concerns in research design, data collection, and analytical techniques. We discuss these next and provide specific recommendations for addressing these concerns in future research (see [Table 5](#)).

Qualitative studies

Assessing methodological rigor of qualitative research is contentious as there are a myriad of positions and perspectives concerning how to conduct qualitative investigations (see [Cornelissen, 2017](#); [Delmar, 2010](#); [Parry, Mumford, Bower, & Watts, 2014](#); [Pratt, Kaplan, & Whittington, 2020](#); [Wright, 2017](#)). However, *The Leadership Quarterly* seeks to publish qualitative research that is reproducible and replicable ([Antonakis et al., 2019](#)). As such, rigorous qualitative studies fully describe what methods have been used to gather data, code, and clearly articulate the reliability of coding protocols and analyses ([Patton, 2002](#); [Wright, 2017](#)). Qualitative researchers should also consider the likelihood of counterfactual interpretation of their collected data ([Geddes, 2003](#); [Gerring & McDermott, 2007](#)). In support of *The Leadership Quarterly's* position, we assessed the thoroughness, transparency, and replicability of all qualitative studies included in this review ($k = 30$), as well as the qualitative studies of included mixed methods articles ($k = 7$, total $k = 37$, 53.62 %), following a procedure recommended by [Aguinis and Solarino \(2019\)](#) that focuses on the transparency and replicability of qualitative research.

Their procedure involved rating each study in terms of its: (1) *research design*, i.e., the qualitative method used, research setting, position of researcher/s along the insider–outsider continuum, sampling procedures, and relative importance of participants/cases; (2) *measurement*, i.e., how participant interactions are documented, theoretical saturation points, recording of unexpected opportunities or challenges, and power imbalance management; (3) *data analysis*, i.e., explanation of first-order and second order coding schemas; and (4) *data disclosure*, i.e., raw data availability. Using [Aguinis and Solarino \(2019\)](#) criteria allowed us to identify problematic areas in qualitative crisis leadership research. We identified the following problematic areas by using this procedure, which are summarized in [Table 5](#), along with suggested remedies.

First, only one study in our review ([Smith et al., 2018](#)) mentioned managing power imbalances between researchers and participants. A power differential can influence the research process and the quality of data gathered in qualitative studies ([Ostrander, 1993](#)). Specifically, the existence of power, domination, and even exploitation dynamics can influence the research process ([Bryant, 2019](#)). For example,

[Smith et al. \(2018\)](#) described the existing power differential between themselves and the two US governors they interviewed by highlighting how interviewing elites may result in revisionist history. To address this power imbalance, the authors used semi-structured interview questions guided by a framework and cross-checked the governors' responses against other sources of evidence, including observations and archives.

Second, we found only one qualitative study ([Bonn & Rundle-Thiele, 2007](#)) that specifically acknowledged unexpected opportunities, challenges, or other events that occurred during data collection. One of the advantages of qualitative research is its openness to theoretical and non-probabilistic sampling, acknowledging that relevant information can emerge from unforeseen opportunities and data sources ([Evered & Louis, 1981](#); [Harvey, 2010](#)). Not all qualitative research faces such unexpected challenges or opportunities, yet if researchers document these events and how they were treated (e.g., attrition, new participants, new data sources), then it provides a richer research context and increases the possibility that such research can provide new insights.

Third, few authors explained the researchers' position along the insider–outsider continuum. It is important to adequately describe existing or non-existent relationships between the researcher and participants because the quality of these relationships can shape the research process (e.g., data access, quality of disclosure) and results (e.g., analysis and interpretation, [Evered & Louis, 1981](#); [Griffith, 1998](#)). For example, in crisis leadership research, it is possible that researchers are also impacted by the crisis. As such, it is critical for them to disclose their status (e.g., victims of the crisis) and to describe the methods they use to achieve objectivity in their analysis (e.g., bracketing, using memos, inter-coder reliability).

Fourth, many of the crisis leadership studies failed to identify theoretical saturation points in their research (e.g., [O'Reilly & Parker, 2013](#); [Saunders et al., 2018](#)). A saturation point is the moment where no new insights or themes can be drawn from continuous data collection ([Bryant & Charmaz, 2007](#); [Strauss & Corbin, 1998](#)). It is essential for scholars to report how they reached saturation, alongside the criteria they used to make this judgment. Only four studies reported how and when saturation was reached ([Mora Cortez & Johnston, 2020](#); [Grover & Hasel, 2015](#); [Hannah & Zatzick, 2008](#); [Varma, 2020](#)); however, they did not fully describe the criteria for making this judgment.

Fifth, many researchers failed to provide a clear and comprehensive description of how they analyzed their data. Specifically, many studies did not provide details of coding stages (e.g., first-order, second-order), higher level data abstraction, or a complete list of codes. Among the articles we reviewed, only three revealed their coding schemas. Specifically, [Liu \(2017\)](#) and [Liu et al. \(2017\)](#) provided accounts of their discourse analysis, along with their data structure, exemplar quotes, and images. Similarly, [Prouska and Psychogios \(2019\)](#) provided a description of their process as well as a list of open codes and key themes.

While [Aguinis and Solarino's \(2019\)](#) rubric provided a framework for assessing the transparency of the qualitative studies included in our review, it does not allow for assessment of qualitative coding reliability, or acknowledge the importance of considering counterfactual evidence, which may render the issues we have mentioned previously moot.¹¹ Qualitative research must contend with debate surrounding sub-

¹¹ We appreciate the feedback of our anonymous methods reviewer on this point.

jectivity and bias in the analysis of collected data (Church, Dunn, & Prokopy, 2019; Olson, Young, & Schultz, 2016). While there are many benefits to coding qualitative data using multiple coders (as we did in this review), a key benefit is the ability to establish both validity and reliability (Church et al., 2019; Morse, Barrett, Mayan, Olson, & Spiers, 2002). Out of the 37 studies that used qualitative data to study crisis leadership, only three (Bligh et al., 2004; Jensen et al., 2021; North et al., 2013) reported inter-coder reliability.

Similarly, considering counterfactual evidence, even as a thought experiment (i.e., imagined; see also Aguinis et al., *in press*), is important in verifying the emerged findings (see Geddes, 2003; Gerring & McDermott, 2007). Counterfactuals are “subjunctive conditionals in which the antecedents of an event are assumed to be known but for purposes of argument supposed to be false.” (p. 64; Jaccard & Jacoby, 2020). In simpler terms, counterfactuals focus on “what-might-have-been” questions (Tetlock & Belkin, 1996) which allows the researcher to arrive at possible alternative explanations and new theoretical insights. Unfortunately, only one of the reviewed qualitative studies incorporated counterfactuals in the analysis (see Swinkels, 2020). Possible remedies for these additional concerns are also reported in Table 5.

Quantitative studies

We also critiqued all the quantitative crisis leadership studies, plus those appearing in mixed methods articles ($k = 39, 56.52\%$).¹² Again, we followed methodological best practice to determine the quality and rigor of these studies (i.e., Antonakis, 2017; Hill et al., 2021). After coding for quality and rigor, we identified several concerns regarding these quantitative studies (see Table 5).

(1) Observational: Archival studies. There were 21 archival data studies (14 archival studies, 7 mixed-methods studies, 30.43 %) that examined crisis leadership by aggregating data from various sources (e.g., firm annual reports, stock market). Although archival data analysis can benefit crisis leadership research, we also noted several concerns. First, researchers constructed their archival dataset by selecting databases mostly from large, well-established publicly traded companies in Western countries (e.g., Gupta, Han, Nanda, & Silveri, 2018; Kashmiri & Brower, 2016; Patel & Cooper, 2014; Sergent & Stajkovic, 2020; Stoker et al., 2019), which limits the generalizability of their research findings. Although some studies utilized fixed-effects models to control for constant extraneous factors (e.g., Gupta et al., 2018; Zhang, Jia, & Gu, 2012), whether the findings from these studies also hold for smaller, private firms, or other countries, needs further empirical validation.

The operationalization of key variables in the archival data studies found in our review is also problematic. Archival studies use proxies instead of direct measures, and therefore use a partial approximation of an underlying construct (Ketchen, Ireland, & Baker, 2013). For example, Gupta et al. (2018) used a 5 % or greater decrease in aggregate industry sales to assess industry shock, yet industry shock may only be partially reflected by a decrease in aggregate industry sales. The discretionary choice of cut-off points reflecting decreases in industry sales is also a concern. The degree to which a proxy reflects the construct can impact the interpretation of study results (Ketchen et al., 2013), and therefore archival data may not be capturing the reality of the crisis accurately. For example, in examining how managers and CEOs dealt with product harm crises, Zhang, Jiang, Magnan, and Su (2021) coded a product harm crisis as a dummy variable. The use of a dummy variable overlooks the fact that the intensity and nature of internal crises can vary across firms. Archival studies should attempt to alleviate this problem by offering a clear justification

for their operational definitions of key variables and acknowledge the limitations of their archival datasets.

In addition to problematic operationalization of variables, there may also be endogeneity concerns due to the use of cross-sectional data which makes it difficult to establish temporal precedence (see Barnes, Dang, Leavitt, Guarana, & Uhlmann, 2018). There are means by which scholars can tackle this issue. For example, in their study of scandal press coverage, Graffin et al. (2013) conducted supplementary analysis to unpack the temporal ordering of the variables by separating the data into two variables based on dates (i.e., pre-scandal coverage and post-scandal coverage). Sajko et al. (2021) took a similar approach in their study of the 2008 GFC, by conducting three sets of analyses that covered different time periods. Further, they used panel data analysis and clustered standard errors within the panel to account for heteroscedasticity and intragroup correlations. In these analyses, predictor variables and controls were all lagged by one year. This analytical design alleviates some endogeneity concerns; however, there are also endogeneity concerns due to omitted variables that may influence both predictor and outcome variables (Barnes et al., 2018; Hamilton & Nickerson, 2003).

Endogeneity is an especially concerning problem in archival research as unmeasured variables may cause spurious correlations (Antonakis et al., 2010; Barnes et al., 2018). All the archival studies in our review included numerous controls to help alleviate this concern. However, the controls included in archival studies are limited by what is available in the databases used by the researcher. Researchers may control for what variables are in their database without a theoretical basis, or they may not be able to control for theoretically relevant variables because they are not available in their database. It would be preferable to use natural experiments to estimate the effects of external crises (which are considered exogenous shocks), but this powerful tool is not often used by management scholars and is considered by many to be a challenging undertaking (see Sieveke & Santoni, 2020). The endogeneity issue of omitted variables was also a concern when analyzing the quality of the survey studies in our review.

(2) Observational: Survey studies. Of the eight survey studies (11.59 %) we reviewed, four were cross-sectional, and four were longitudinal designs. While observational studies using surveys are commonly employed in leadership research, they are often prone to endogeneity problems (see Antonakis et al., 2010). Unsurprisingly, some common causes of endogeneity (e.g., omitted variables, simultaneity, measurement error, common-method variance, and selection, Antonakis et al., 2010; Hill et al., 2021) were seen in these eight studies too.

While some researchers attempted to address the omitted variables issue by including demographic variables such as gender, age, and educational level, scholars seldom provide theoretical or empirical justification for their choice of control variables (as recommended by Bernerth, Cole, Taylor, & Walker, 2018). Conversely, other important control variables (other than demographic variables) were often missing. Social desirability, for example, is often an important control variable in organizational research (Bernerth & Aguinis, 2016), yet we found only two articles that included social desirability as a control variable (Williams, Pillai, Lowe, Jung, & Herst, 2009; 2012). However, it is important to note that the use of such control variables alone does not resolve the issue of endogeneity. Endogeneity can only be addressed in this scenario by using instrumental variables that rely on exogenous sources of variance (Antonakis et al., 2010; Hill et al., 2021; Sajons, 2020); yet these are a challenge to find and were not typically employed in the studies we reviewed.

Common method variance was also a problem with many of these studies. For example, Boehm et al. (2010) collected data from a single source at one time point without applying any statistical procedure to address the issue of common method variance. Other studies estimated common method variance using Harman's one-factor test, which is generally viewed as unsuitable for this purpose (see Antonakis et al.,

¹² Mixed method studies were counted here in the analysis of both qualitative and quantitative studies, so the number of studies does not add up to 69.

2010; Richardson, Simmering, & Sturman, 2009). Only a small number of studies either collected data from multiple sources (e.g., Akbar & Aldrich, 2015; Davis & Gardner, 2012; Sommer et al., 2016) or collected data at different time points (e.g., Williams et al., 2009; 2012) as a way to address common method variance. We note that while these are possible remedies for common method variance concerns, they do not correct for omitted variables and endogeneity remains a problem.

Measurement error is another threat to validity that was common in the survey studies found in our review. The most common approach to measurement by scholars was to use popular scales that report adequate reliability in the scholarly literature, e.g., Sommer et al. (2016) used the Multifactor Leadership Questionnaire (MLQ) Form 5X-Short to evaluate transformational and transactional forms of leadership (Avolio & Bass, 1995), while Williams, Pillai, Deptula, and Lowe (2012) used the measure of authentic leadership developed by Walumbwa, Avolio, Gardner, Wernsing, and Peterson (2008). On a positive note, most of the survey studies reported alpha reliabilities for their scales. However, in some cases there were questions over scale validity. There were cases where studies utilized scales developed in another language and translated them for use in another country, but they did not demonstrate cross-cultural measurement equivalence (Davidov, Meuleman, Cieciuch, Schmidt, & Billiet, 2014). This is problematic as the wording of scale items may not mean the same thing in different cultural contexts, leading to measurement errors and scale invalidity. Notably, none of the survey studies in our review explicitly modeled measurement error (Bollen, 1989; Culpepper, 2012).

Simultaneity is another cause of endogeneity problems in survey studies, especially cross-sectional studies (Antonakis et al., 2010; Hill et al., 2021). Even in studies where data were collected at multiple points in time, causality is not guaranteed, as temporal order is a necessary but insufficient condition (Antonakis et al., 2010; Kerlinger & Lee, 2000). We did not see attempts in the studies we reviewed to address simultaneity issues with recommended procedures, such as instrumental variables (Hill et al., 2021). This is likely because instrumental variables can be difficult to find (Hill et al., 2021; Sajons, 2020). The most promising way for scholars to establish causality is the use of experimental designs (Antonakis et al., 2010), although this approach to crisis leadership research presents its own challenges.

(3) Experimental vignette studies. Experimental vignettes feature in 12 of the quantitative studies (17.93 %). Given the rarity of crisis events and the difficulties in collecting field data during a crisis (Morton & Levy, 2011), experimental vignettes are an attractive option for researchers. Such methods are also a relatively easy and safe way to manipulate independent variables in crisis leadership research. However, the experimental vignette studies in our review suffered from a range of methodological concerns.

First, vignette experiments have been criticized as unrealistic, and results often lack ecological validity (Hughes & Huby, 2002). Despite calls to make vignettes more immersive and realistic (Aguinis & Bradley, 2014), most studies in our review used written vignettes with only one study utilizing more realistic video vignettes (Verčič et al., 2019). Immersion is an important design consideration since realistic perceptions of a crisis experience is crucial for detecting responses to such events outside the laboratory. However, this can be difficult to achieve in practice. For example, some of the experimental studies excluded from our review claimed to study a crisis, but instead simply manipulated time pressure to complete a task (Pillai & Meindl, 1998), or did not study an organizational crisis according to our definition (see also Koehler & Raithe, 2018; Spector, 2020).

Second, we did not find a single vignette study that measured a behavioral outcome. The outcomes in all studies were hypothetical and were non-costly to participants, which can increase socially desirable responding (see Lonati, Quiroga, Zehnder, & Antonakis, 2018). For example, some studies measured participants' future purchase

intentions following a product harm crisis (e.g., Cowen & Montgomery, 2020; Laufer, Garrett, & Ning, 2018), but intentions do not have the same consequences as behaviors for participants. To increase the external validity of vignette studies, where possible, researchers should incorporate outcomes that capture actual participant behaviors (Aguinis & Bradley, 2014). For example, a vignette study about responses to crises caused by leader scandals (e.g., fraud, sex scandals) could incorporate the voodoo doll task (VDT). This task of sticking pins into a voodoo doll has been validated as a way of measuring harming behaviors (DeWall et al., 2013; McCarthy, Crouch, Basham, Milner, & Skowronski, 2016) and has already been used in leadership research (Liang et al., 2018). Another way to address the absence of a behavioral outcome, is to utilize multiple studies of different designs. For example, Stam et al. (2018) used archival data, a lab experiment, and a vignette experiment. In the lab experiment, the authors measured actual task performance of the participants. Incorporating multiple studies of different designs provided a means to combat issues with both realism and the lack of behavioral outcomes.

Third, very few studies conducted a priori power analysis to determine the desired sample size or reported the ad-hoc power (exceptions include Cowen & Montgomery, 2020; Jensen et al., 2021). Such practices inform others about the confidence of detecting the true effect. All but one study (Jensen et al., 2021) used ANOVA, ANCOVA, or MANOVA as the statistical procedure analyzing main effects. These are valid methods of statistical analysis, yet the studies we reviewed rarely incorporated control variables. Even if randomization in a vignette experiment works, failing to include appropriate control variables in an experimental study may bias the estimated effects (Antonakis et al., 2010). This is because including covariates increases precision of the estimated treatment effect and is particularly important to reduce bias if groups are not perfectly balanced.

Finally, we found only one article (Jensen et al., 2021) that explicitly addressed endogeneity by using an instrumental-variable procedure (see Sajons, 2020). In Jensen et al. (2021), leader charisma was manipulated, and the instrumental variable procedure was used to examine the effects of beliefs (of what others would do) on physical distancing (the outcome variable). Next, we provide several suggestions for improving the methodological rigor and quality of future crisis leadership research.

Improving the quality of future crisis leadership research

The qualitative and quantitative methodologies used in the crisis leadership studies we reviewed also allow us to provide targeted recommendations for future research. We identified several concerns regarding replicability and transparency in the qualitative research, and endogeneity in the quantitative research on crisis leadership. In Table 2 and 5, we outlined the criteria set forth by methodological experts in our field to promote high quality qualitative and quantitative research. We reported our evaluation of the literature and based on this analysis we outlined specific recommendations in Table 5 to guide future crisis leadership research. In this section, we provide some suggestions for improving crisis leadership research, focusing more broadly on research design.

Sampling challenges. Studying leadership before, during, and after a real-world crisis is challenging. As crises are (by definition) sudden unexpected events, it is very difficult to plan and conduct high quality research on them. Organizations may be too busy responding to a crisis, or too sensitive to the reputational damage arising from a crisis (particularly if it is poorly managed), to permit access to outsiders (Pearson & Clair, 1998). Similarly, those who have been directly affected by the crisis may find it difficult to participate in data collection, as it may aggravate their trauma and sense of loss. For example, in cases where internal-intentional crises are caused by organizational leaders (e.g., fraud), employees may feel uncomfortable participating in research asking about their leaders' immoral or unethical behavior.

As such, crisis leadership researchers need to acknowledge that their prospective research participants may be vulnerable.

In recognition of the unique challenges in data gathering among those who experience various crises, crisis leadership researchers may benefit from scholars who have conducted research among members of vulnerable populations. For example, [Ellard-Gray, Jeffrey, Choubak, and Crann \(2015\)](#) emphasized the following aspects when studying vulnerable individuals: (1) using the right language from point of invitation to actual interactions, (2) building rapport and respecting participant's autonomy and right to withdraw, (3) reducing sampling bias through mindful efforts to obtaining a representative sample when no sampling frame exists (e.g., time-space sampling), and (4) considering the use of screening surveys to identify a hard-to-reach or hidden sample (e.g., victims of leader abuse) before actual data collection. While there are risks of magnifying trauma by asking crisis victims to recall their experiences, research participants can also experience a catharsis where involvement helps them process and derive new meaning from their negative experiences (see [Bourne & Robson, 2015](#); [Smith et al., 2018](#)).

An additional difficulty researchers may face when investigating crisis leadership is obtaining ethics approval to conduct crisis-related research. This may prove to be a challenge for many scholars given that ethics boards are extra sensitive in evaluating research proposals that involve vulnerable populations. [Restubog et al. \(2021\)](#) offer practical tips on preparing ethics protocols for studies that involve sampling from these types of participants including clearly establishing *beneficence* (i.e., any benefits that outweigh possible risks; [Anderson & Hatton, 2000](#)), *non-maleficence* (i.e., avoidance of unnecessary harm; [Schwartz, Dunfee, & Kline, 2005](#)), and *informed consent* (i.e., the process of informing participants about the potential risks of a research study and obtaining their agreement to take part in the study; [Perrault & Keating, 2018](#)).

Crisis as a process, not an event. Our review found that many of the quantitative studies on crisis leadership focused on a single crisis event at one point in time using cross-sectional field surveys or by conducting vignette experiments. While these studies can provide a snapshot in time and allow comparisons to be made (e.g., between trust repair strategies), they are somewhat unrealistic in the sense that they ignore leadership before and after a crisis event. In contrast, the qualitative studies in our review examined crisis leadership at various stages of the crisis lifecycle. This is a more realistic approach, as a real-world crisis is not a static event, but an evolving process involving pre-crisis, crisis, and post-crisis stages and different leader responses (see [Bavik et al., 2021](#); [Boin et al., 2005](#); [Bundy et al., 2017](#); [Wooten & James, 2008](#)).

As such, future quantitative researchers investigating crisis leadership should consider employing longitudinal designs (e.g., cross-lagged modelling, latent growth modelling, and latent change score modelling, [Liu, Mo, Song, & Wang, 2016](#); [McFarland, Reeves, Porr, & Ployhart, 2020](#)). These methods are particularly suited to examining the dynamics of crisis leadership during the different phases of an external crisis, such as an earthquake. However, scholars need to make sure that the event/treatment is exogenous to the research question; otherwise, the estimated coefficients would be biased.¹³ The use of longitudinal designs may also allow scholars to learn more about the stages of the crisis lifecycle that were underrepresented in our review. These include crisis planning ([Liu et al., 2021](#)), decision-making, crisis coordination ([Mora Cortez & Johnston, 2020](#)), recovering from a crisis ([North et al., 2013](#)), and learning from a crisis ([Hede, 2011](#)).

A crisis might also be modeled as a natural experiment. Some of the real crises studied in this review can be considered as exogenous shocks (for external crises) and could be examined using quasi-experimental procedures, such as regression discontinuity in time

(e.g., [Hausman & Rapson, 2018](#)) or difference-in-difference comparisons (to contrast organizations and/or leaders affected vs not affected by a crisis, see [Sieweke & Santoni, 2020](#)). The one example of a natural experiment in our review is provided by [Birkeland et al. \(2017\)](#) who compared employees directly exposed to a terror attack with a matched group of employees who were not. Their design enables a causal relationship to be inferred, as the treatment (the exposure to the terror attack) can be considered random, and the employees in the treatment group and control group were matched on demographic characteristics. Despite the inherent challenges in using such designs, this approach to data collection yields valuable insights.

Consider different levels of analysis. In addition to being focused on a single point in time, most of the quantitative research in our review focused on a single level of analysis (i.e., individual level perceptions of crisis leadership, or organization level responses). However, a leader's influence extends within the organization, to teams and individuals, and beyond the organization to external stakeholders, and vice versa ([Wo, Schminke, & Ambrose, 2019](#)). Multilevel research could disentangle the unique consequences of crisis leadership across the individual, team, and organizational levels, and consider higher-level boundary conditions that might mitigate or exacerbate the impact of crisis leadership. We encourage crisis leadership scholars to move beyond individual level analyses, to consider crisis leadership at a dyadic or team-level. Employing social network analysis, for example, could allow the integration of multiple levels of analysis and enable a simultaneous examination of the whole, as well as the parts ([Kilduff & Brass, 2010](#)). In [Table 4](#), we present some ideas for multi-level studies in each quadrant of our crisis leadership typology.

Use an internal perspective. In our review of the crisis leadership literature, we found no studies examining crisis leadership from a within-person perspective. However, within-person experience sampling offers researchers the ability to capture "life as it is lived" ([Bolger, Davis, & Rafaeli, 2003](#)), and we argue that this approach could provide unique insights into crisis leadership (see also [Fisher & To, 2012](#)). Depending on the type of crisis, participants could report their experiences at pre-determined intervals (i.e., interval-contingent design, e.g., measuring individuals' wellbeing in the morning, afternoon and evening during a crisis), or self-reported each time a particular event in question occurs during and post crisis (e.g., measuring individuals' responses to each email they receive from their leader). Capturing stakeholders' perceptions of crisis leadership in this manner would be interesting, but we argue that this approach would be even more valuable if it were used to capture the lived experiences of the leader themselves during a crisis in real time.

Despite being published in high quality journals, the empirical evidence discovered in this review suggests that there is room for improvement in the methods used to investigate crisis leadership. Much of the qualitative research draws on case studies or interviews that suffer from biases and lack generalizability. Additionally, most of the quantitative studies only provide snapshots in time that utilize cross-sectional surveys or unsophisticated experiments, each with a myriad of endogeneity issues. Furthermore, many studies only examine one level of analysis as opposed to bridging different levels of analyses (see [Pinto, Leana, & Pil, 2008](#)). In sum, crisis leadership scholars need to increase the rigor and quality of their research designs and data analyses to improve the contribution of future crisis leadership research.

Conclusion

Our systematic interdisciplinary review of the crisis leadership literature contributes to our understanding of leadership during crises in at least three important ways. First, by utilizing the [Coombs and Holladay \(1996\)](#) crisis typology, we bring much needed clarity to understanding the way that leaders respond to different crisis types,

¹³ We thank our anonymous methods reviewer for this suggestion.

and the outcomes of these responses on internal and external stakeholders. In this sense, our review identifies important leadership themes emerging from the literature that are reflected within and between different types of crises. Second, we provide a critique of the qualitative and quantitative methods currently used to examine leadership during crises. This provides a general quality assessment of the conclusions that can be drawn from the current literature, as well as specific methodological issues that require close attention in future research. Finally, our review provides management scholars with exciting suggestions for future crisis leadership research, and practical guidance for leaders to successfully respond to different types of crises. Given the forecast for unpredictable storms and rough seas, we hope that our crisis leadership review and methodological critique provides valuable guidance for those at the helm of leadership theory and practice.

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*References marked with an asterisk indicate the empirical studies included in our review.

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